

MODELVIEWER 2026.11

Simulation Results and 3D Plotting

A user guide to reading solver results and plotting your own data in ModelViewer: from opening a file to a finished report.

User guide · for ModelViewer 2026.11

About this guide

ModelViewer began as a viewer for 3D models. Two of its newer areas go beyond models: it shows the **results of engineering simulations** that were calculated elsewhere, and it **plots your own data in 3D**. This guide covers both in depth, and shows how they work together. The tutorial lessons (Help → Tutorial, lessons 19 to 25) walk through the same ground step by step; this guide is the reference you keep open next to the program.

> **WHAT MODELVIEWER DOES AND DOES NOT DO**
> ModelViewer **displays** the results that a solver has written. It does not mesh, solve or run simulations, it does not edit results, and it has no scripting or expression language for post-processing. Its strengths are the ones a viewer needs: fast, correct colours; units; time; sections and probes; side-by-side comparison; and the presentation tools (measurement, annotation, PDF report, image export) that the rest of the program offers.

Who it is for

Engineers who read the output of structural, thermal or flow analyses (CalculiX, Code_Aster, Elmer, FreeCAD FEM, OpenFOAM, MOOSE and similar open solvers) and want to inspect, compare and present it; and anyone who has tables of numbers - measurements, simulation exports, mathematical functions - and wants to see them as surfaces, curves, points, bars, volumes or vector fields next to a 3D model.

How the guide is organised

Part	Content
I – Simulation results	Opening a result, fields and colours, units, deformation, time, looking inside a solid, probing and charts, comparing, saving and large results (chapters 1 to 9).
II – 3D plotting	Plots from CSV files and from formulas, the plot types, flow in time, styling and shared axes, notes, images, editing and export (chapters 10 to 17).
III – Together	Results and plots on one clock, and worked workflows (chapters 18 and 19).
Appendices	Catalogue of the sample files, a quick reference, messages and troubleshooting, glossary.

Conventions

- Menu paths use arrows: **File** → **Open...**. Names of controls, tabs and buttons are in **bold**, spelled as in the English interface.
- Keys look like <kbd>Ctrl</kbd>+<kbd>O</kbd>. File names and column names are in `monospace`.
- Boxes marked **Note** add background, **Tip** a shortcut or good habit, **Caution** something that surprises people or a limit of the program.

- All sample files named here ship with ModelViewer in the `sample-models` folder (`Simulation` and `Plot3D`); Appendix A lists them. The pictures were taken with the English interface.

> **TIP**
> Everything here is also in the program: **Help** → **What's New…** gives an overview of the 2026.11 release, **Help** → **Quick Help** has tabs for Simulation Results, 3D Plots and Analysis, and hovering over a control shows a short explanation.

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PART I

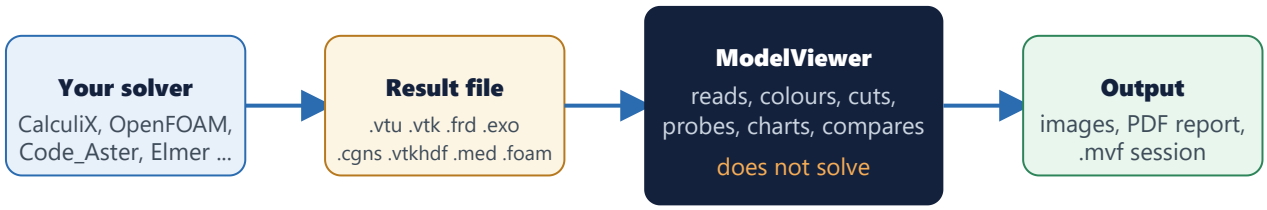
Simulation results

1 Introduction to simulation results

A simulation - a finite-element stress analysis, a heat-transfer run, a flow calculation - ends with a result file: the mesh the solver worked on, and for each point or cell of it one or more **fields** (stress, displacement, temperature, velocity, pressure ...), usually for several **steps** (load steps, vibration modes, moments in time). ModelViewer reads such a file and shows the model coloured by a field. This chapter explains the idea and the vocabulary; the following chapters go through the features one at a time.

1.1 A viewer, not a solver

The boundary matters, because it explains both the strengths and the limits. ModelViewer is **read-only** with respect to the physics: it never changes a value, never runs a calculation of its own other than the display arithmetic described below (derived stress values, unit conversion, averaging), and has no solver, mesher or expression language. In exchange it stays small and fast, and every result gets the same presentation tools as any model: selection, sections, measurement, annotation, report export and the rest of the viewport.



Where ModelViewer sits in a simulation workflow

1.2 What you see: the boundary surface

A solid result has millions of cells inside, but you look at a solid from outside. ModelViewer therefore extracts the **boundary surface** - the faces that belong to exactly one cell - and draws that, coloured by the chosen field. The interior is not lost: sections, iso-surfaces, streamlines and volume rendering (chapter 6) reach into it, and probes and charts read values from it. Results that already are surfaces (shell models made of triangles or quadrilaterals) are drawn as they are.

- Supported cell types: tetrahedra, hexahedra, wedges, pyramids, their quadratic forms (drawn through their corner nodes) and polyhedra with explicit faces. Triangles and quadrilaterals are shells.
- Cell types that have no surface to draw - polygons, triangle strips and single vertices of the legacy VTK format, and the 1-D beams of a network - are read but not displayed; this is expected.
- The displayed surface is an ordinary mesh in the scene, so it can be selected, hidden, transformed, measured and annotated like any other.

1.3 Vocabulary

Field
: One kind of value over the model: a scalar (temperature), a vector (displacement, velocity - three components) or a tensor (stress - six components).

Node data and cell data
: A **node** field has one value per mesh point and is drawn smoothly interpolated. A **cell** field has one value per element and is drawn in one flat colour per element (marked **[cells]** in the field list).

Step
: One state of the result: a load increment, a time step, or a vibration mode. A result with several steps shows a timeline.

Component
: One of the parts of a vector or tensor (X, Y, Z ...) or its magnitude.

Range
: The values that the colour scale spans, from its minimum colour to its maximum colour.

Derived field
: A value computed from another one at load time - for stress files, von Mises stress, Tresca stress, the three principal stresses and the maximum shear.

1.4 Supported formats

The readers are ModelViewer's own; there is no dependency on VTK or ParaView. Which formats appear in the Open dialog depends on the libraries your build includes.

Format	Extensions	Notes
VTK	`.vtu` , `.vtk`	The common interchange format. XML files (inline, appended raw and compressed data) and legacy files (ASCII and binary). Polyhedra are supported. FreeCAD's FEM export writes `.vtu` .
CalculiX	`.frd`	The result file CalculiX writes (and FreeCAD FEM keeps). Displacements, stresses and strains; von Mises, principal stresses and maximum shear are derived while loading. Values are in the solver's own units.
Exodus II	`.e` `.exo` `.ex2` `.g`	Common for MOOSE and Sierra results. Needs a build with NetCDF. Node and element variables, many steps.
CGNS	`.cgns`	Unstructured and structured zones. Needs a build with the CGNS library. Vertex and cell-centre solutions; boundary-condition sections are skipped.
VTKHDF	`.vtkhdf`	The HDF5-based VTK format as ParaView writes it, including time steps. Unstructured grids, poly data and image data are read; composite files are not supported yet. Needs HDF5.

Format	Extensions	Notes
MED	`.med`	Code_Aster and Salome results (MED 3 and later). Needs HDF5. Older Code_Aster 2.x-layout files must be converted (Salome's *medimport*).
OpenFOAM	`.foam`	Open the (empty) `.foam` file in the case folder. The case's time directories become the steps. Polyhedral meshes are supported.

> **NOTE**
> **Help** → **About** lists the optional libraries of your build. If an extension is missing from the Open dialog's filter, that build was made without the matching library, and the format will not open.

1.5 The workflow in short

1. Run your analysis in the solver of your choice and keep the result file.
2. Open it in ModelViewer (**File** → **Open...**), pick a field, check the units.
3. Look: rotate, section, step through time, probe values, chart them, compare with another run.
4. Present: add annotations or measurements, capture views, export a PDF report or an image, or save a `.mvf` session that contains the result.

2 Opening a result

2.1 Two ways to open

Command	What happens
File → **Open...** (or Ctrl + O, or drag the file onto the window)	The result opens in a **document of its own**. This is the normal way and the one to use first.
Visualization → **Simulation Results** → **Add Result to This Document...** (also **Add Result...** in the tab)	The result is added to the **current document**, next to whatever is there - a CAD model, another result. Use it to compare two results (chapter 8) or to see a result together with a plot (chapter 18).

Both accept the same files. The file filter offers *All Simulation Results* and one entry per format. An OpenFOAM case opens through the empty `.foam` file in the case folder.

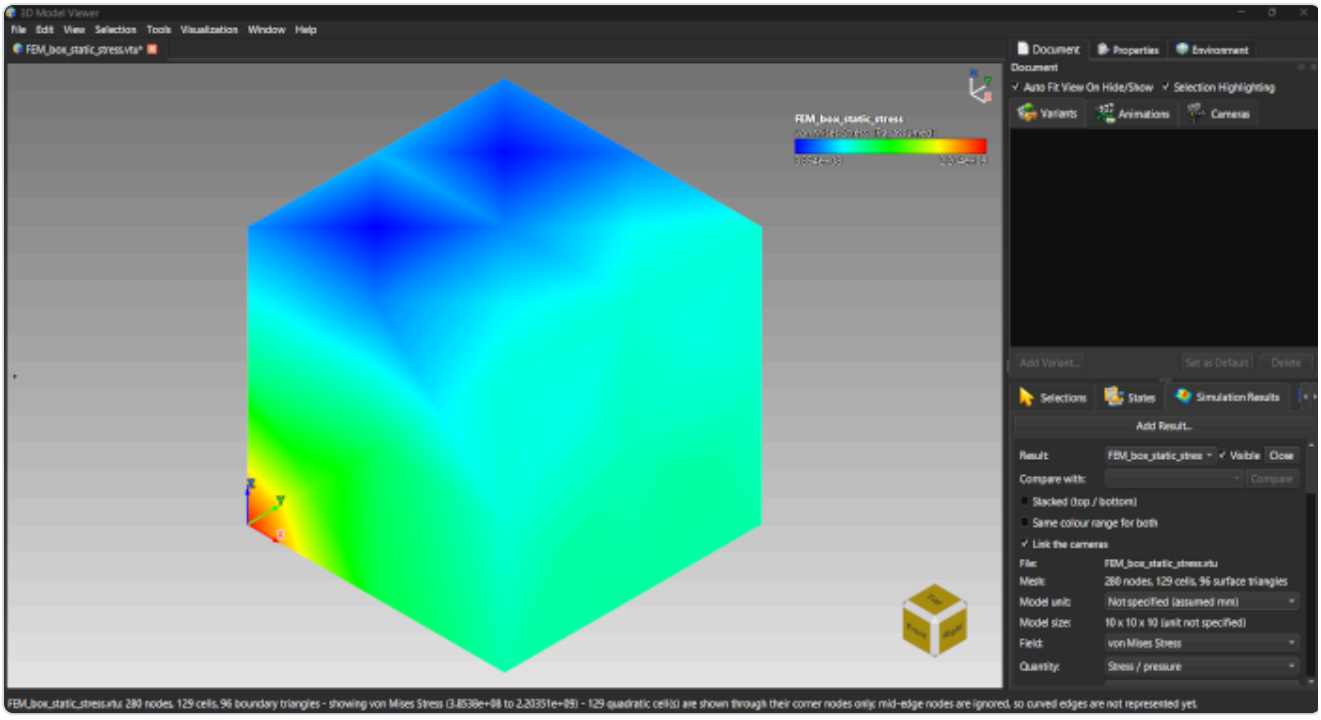
2.2 While it loads

Reading, decompressing, tessellating and extracting the boundary surface run in the background, with a progress bar and a Cancel button, so the program stays usable. Only one result loads at a time; starting a second while one is loading gives a message. Large text formats (ASCII `.frd` or `.vtk`) are much slower to read than the binary HDF5 and NetCDF based ones, which is a property of the formats, not of the program.

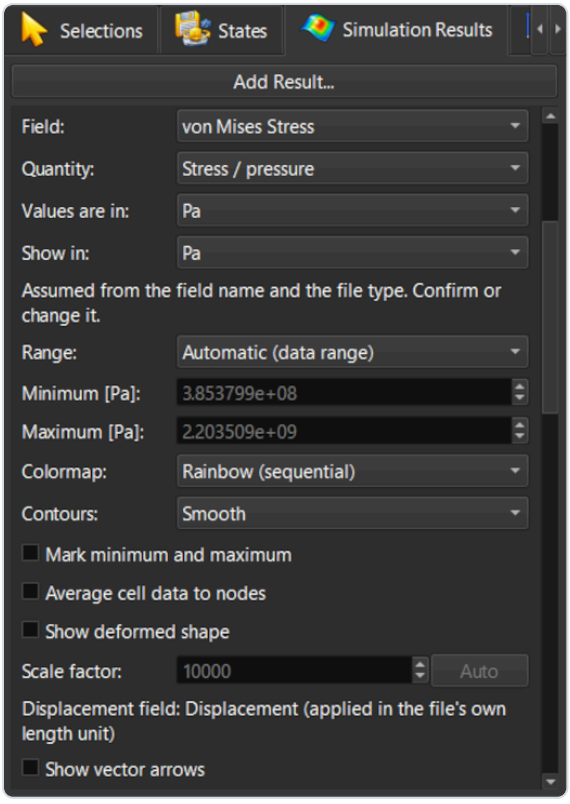
2.3 The Simulation Results tab

Once a result is loaded, the dock at the bottom left switches to the **Simulation Results** tab. It holds nearly everything in this part of the guide, and its controls are inactive until a result is selected. From the top:

- **Result**: which result of the document the controls act on, with a button that removes the selected result from the document (this can be undone), and an info line with the file name and the counts - nodes, cells and surface triangles.
- **Model unit** and **Model size**: the length unit of the geometry (metres, centimetres, millimetres, inches, feet; when the file does not say, *Not specified (assumed mm)*) and the size of the model in it.
- **Field**, **Component**, **Quantity** and **Values are in**: what to colour by, and its unit (chapter 3).
- **Range**, **Colormap**, **Contours**, the deformation and marker options, and the groups for sections, iso-surfaces, streamlines, volume, arrows and ellipsoids (chapters 3, 4 and 6).
- **Charts**: the buttons for plotting over a line or over time and for the distribution (chapter 7), and the **Compare** controls (chapter 8).



A result just opened: the box coloured by von Mises stress, with its legend, and the Simulation Results tab at the bottom left



The Simulation Results tab: field, component, range, colormap and contours

2.4 Results without fields or without volume

- A file that contains only geometry shows the geometry, and the tab says so: there are no values to colour by.

- A shell result (triangles and quadrilaterals only) has no volume. Everything that needs a volume - sections that reveal the inside, iso-surfaces, streamlines, volume rendering - reports that there is nothing to cut or trace. Colours, deformation, probes and charts all work.
- A result without a displacement field cannot be shown deformed; the option explains this.

> **TIP**
> If you are unsure what a file contains, open it and look at the **Field** list and the info line first: they show every field the reader found, which steps exist, and how many nodes and cells it has. For a first try, `sample-models/Simulation/FEM_box_static_stress.vtu` is small and opens coloured by von Mises stress.

3 Fields, colours and units

3.1 Choosing the field

Field lists everything the file contains, plus the derived values. A vector field offers **Component** choices - its components and its *magnitude*; the six components of a stress tensor are summarised by the derived scalars described below. Fields marked **[cells]** are element data.

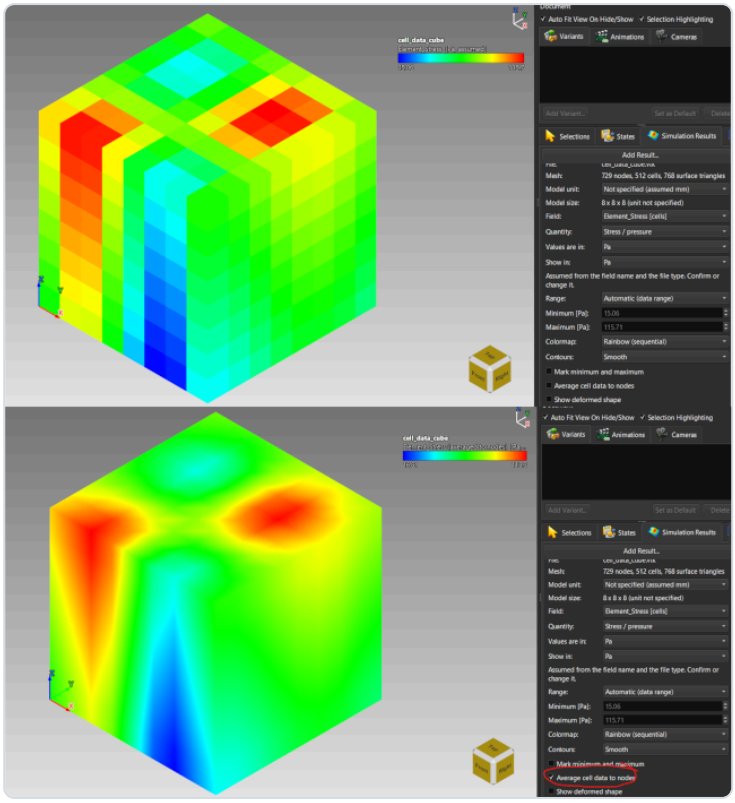
Derived stress values

When a file contains a symmetric stress tensor (six components, in the order XX, YY, ZZ, XY, YZ, ZX), ModelViewer derives the quantities engineers usually want and adds them to the field list as ordinary scalars: **von Mises** stress, **Tresca** stress, the three **principal stresses** and the **maximum shear**. They are computed per node (or cell) from the tensor as each step is loaded; they inherit the unit of the stress and are not stored in a saved session, because they can always be recomputed. CalculiX `.frd` files are the typical source; FreeCAD's `.vtu` export already contains some of them.

3.2 Node data and cell data

A **node field** has a value at every mesh point and is interpolated smoothly over each face, so a coarse mesh still shows smooth colour. A **cell field** has one value per element: each element is drawn in a single flat colour, which is the honest picture of element-wise data (for example an element quality, or most OpenFOAM fields) but looks blocky.

Average cell data to nodes (enabled while a cell field is shown) draws the field smoothly instead: each node takes the average of the elements around it, weighted by their size, so a small element counts for less than a large one. It is a display choice - the data itself is not altered - and it is also what charts, volume rendering and iso-surfaces use for a cell field. Leave it off for labels such as a material or group number, whose regions are identifiers, not quantities.



A cell field drawn flat (left) and averaged to the nodes (right)

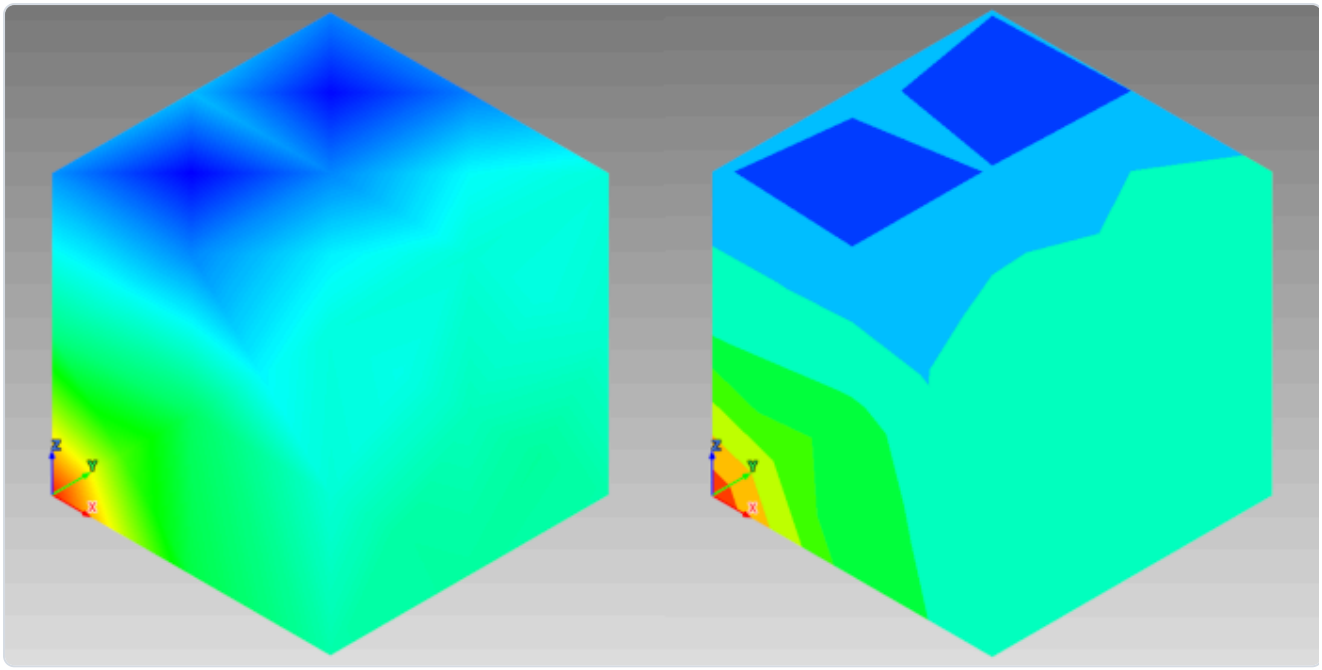
3.3 The colour scale

Range

Range setting	The scale spans …
Automatic (all steps)	the smallest and largest value of the field over **all steps**. This is the default: it keeps the frames of an animation comparable, because a colour means the same value in every step.
Automatic (this step)	the smallest and largest value of the **current step** only. Every frame uses its full colour range, which shows structure well but makes frames incomparable.
Custom	a **Minimum** and **Maximum** that you type, in the display unit. Values outside are shown in the end colours. Use it to compare runs or to enlarge an interesting band.

Colormap, bands and legend

Colormap chooses the ramp: sequential ramps such as the rainbow, and a diverging blue-white-red ramp that suits quantities with a meaningful zero (a displacement component, a signed stress). **Contours** turns the smooth gradient into a set of discrete bands (or back); with bands, the colour boundaries lie exactly at equal values and inside the triangles, not at mesh edges. The legend beside the model shows the scale, its unit and the step; result surfaces are drawn unlit, so lighting never alters the colours.



Smooth colours (left) and the same field with contour bands (right)

Min / max markers and the hover readout

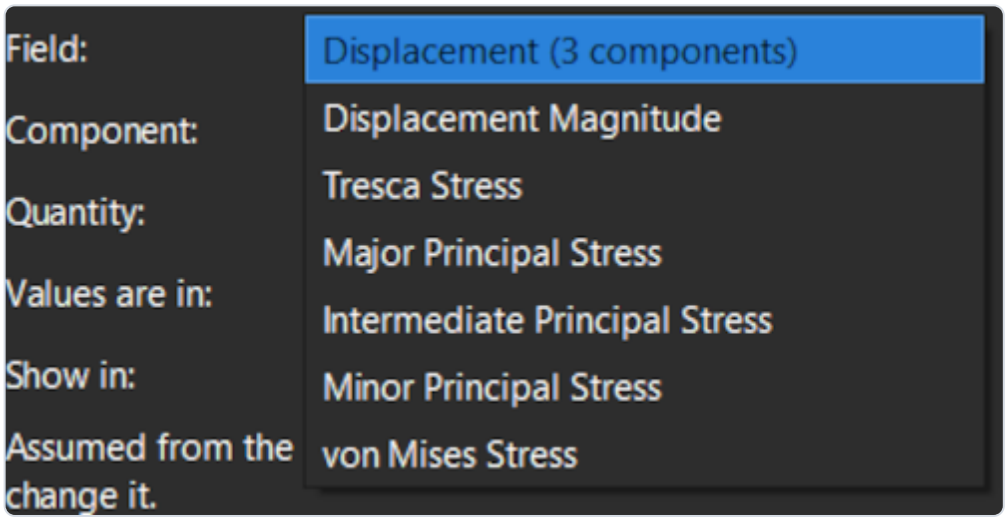
Mark minimum and maximum labels the lowest and highest values *on the visible surface* (the true extremes of a solid may lie inside; sections reveal them). Independently, **hovering** over the model with the mouse shows the value of the shown field at the cursor, with its unit. The hover works whenever a result is selected, except while two results are being compared (chapter 8).

3.4 Units

A wrong unit on a stress or a temperature is exactly the error a presentation tool must not make, so units are explicit and never guessed silently.

- Every field has a **quantity** (stress, displacement, temperature, velocity, pressure ...) and two units: the one in the *file* and the one *displayed*.
- **Quantity** and **Values are in** let you state them. When the file carries a hint, ModelViewer uses it (OpenFOAM states its units through the dimensions of each field: temperature in K, velocity in m/s, a stress tensor in Pa; a kinematic pressure has no unit); when it does not, you choose, and the panel marks the unit as assumed until you confirm it.
- Conversions are more than scale factors: temperature moves between kelvin, degrees Celsius and degrees Fahrenheit, which is an offset as well as a factor.
- The legend, the hover readout, the markers, the charts and the report all use the displayed unit.
- Derived fields inherit the unit of the field they come from.

> **CAUTION**
> Solvers write what you gave them. A CalculiX run with a model in millimetres and Young's modulus in MPa reports stress in MPa and displacements in mm, but the file does not say so - ModelViewer shows plain numbers until you choose the units (the sample `FEM_box_static.frd` is in mm and MPa; its `.vtu` sibling shows Pa). Always check **Values are in** against the units of your analysis.



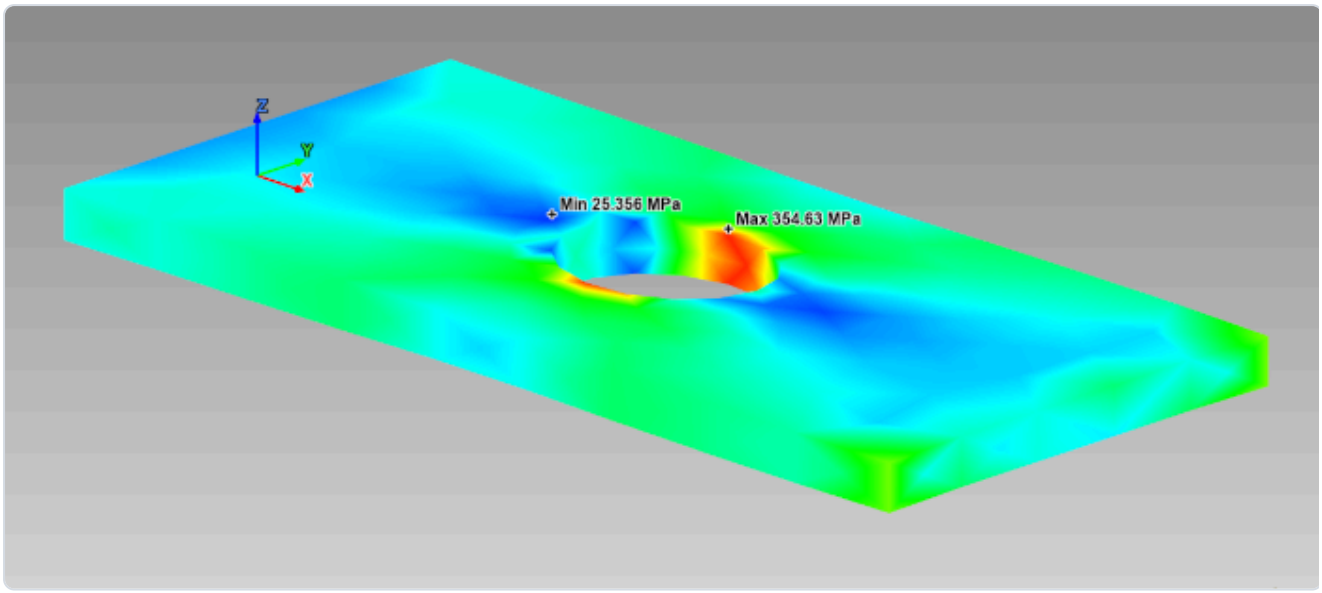
The field list of the static box result

4 The deformed shape and modal results

4.1 Showing the deformation

Structural results contain a displacement for every node. Tick **Show deformed shape** and the model is drawn displaced by that field instead of at rest. The displacement is applied as the model is drawn; the geometry in the file and the rest positions are not altered, so you can switch the option off again at any time.

- **Scale factor** exaggerates the displacement. Real deflections are usually tiny compared with the model, so a factor well above 1 is normal. The **Auto** button picks a factor that makes the movement clearly visible.
- The displacement is applied in the file's own length unit; the panel names the field that is used. A result without a displacement field cannot be shown deformed, and says so.
- While the deformed shape is on, everything that follows the geometry follows the displaced mesh: the hover readout, the extrema markers, sections, iso-surfaces and streamlines.



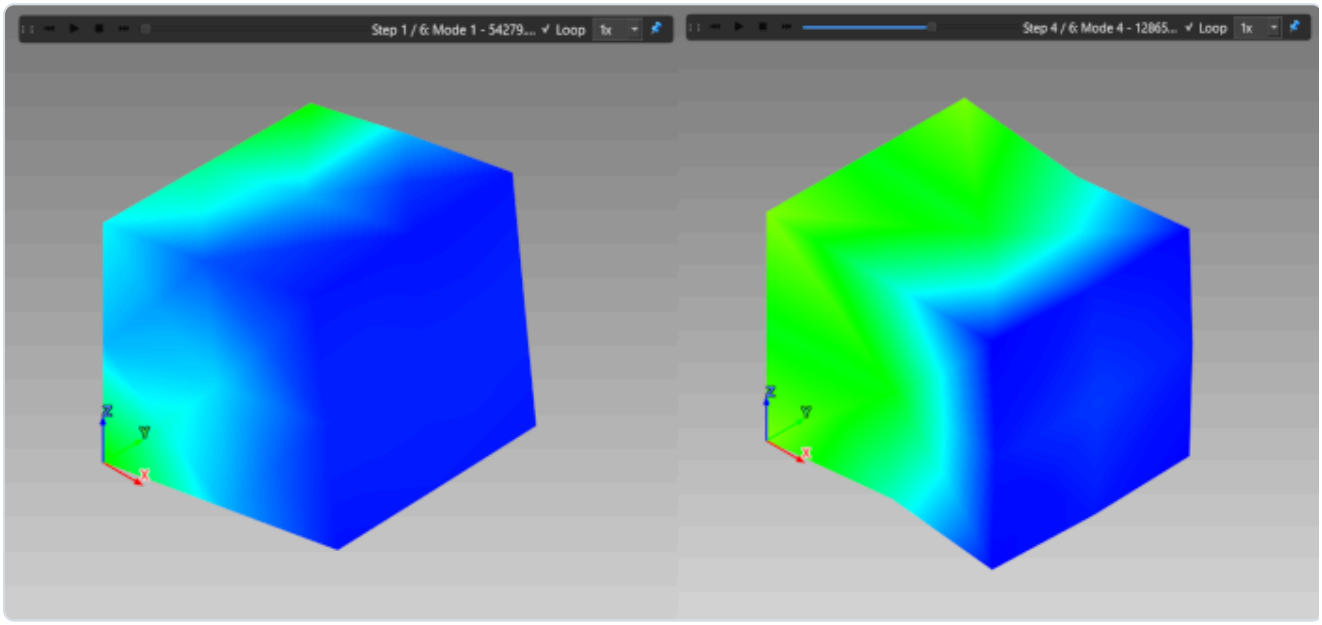
A plate with a hole under load: deformed shape on, minimum and maximum marked

> **CAUTION**
> The picture is exaggerated on purpose. When you judge how much a part really moves, read the scale factor, and use the hover readout or a chart for the true displacement values.

4.2 Modal results

A frequency analysis produces **mode shapes**, not displacements with a physical size: a mode only says how the structure moves relative to itself, and its amplitude is arbitrary. ModelViewer treats such results (steps labelled *Mode 1*, *Mode 2* ... with their frequencies in hertz) specially:

- Each mode is drawn with its largest displacement at a tenth of the model size, so every mode is visible without fiddling; the **Scale factor** scales that (a factor of 1 means a tenth of the model size).
- The steps are the modes. The timeline steps through them and plays them one after another; the label shows the mode's frequency.
- The x axis of a chart of a modal result is the frequency of each mode, which is what makes a frequency-response curve line up with the modes (chapter 7).



Two vibration modes of the box, drawn deformed

> **NOTE**
>
> Samples to try: `FEM_box_modes.frd` (six modes) and `FEM_box_frequency_mode1.vtu` (the first mode, as written by FreeCAD). `FEM_box_static_stress.vtu` and `plate_with_hole.frd` are good for the ordinary deformed shape; `plate_with_hole.frd` has four load steps.

5 Time: steps and the timeline

5.1 What a step is

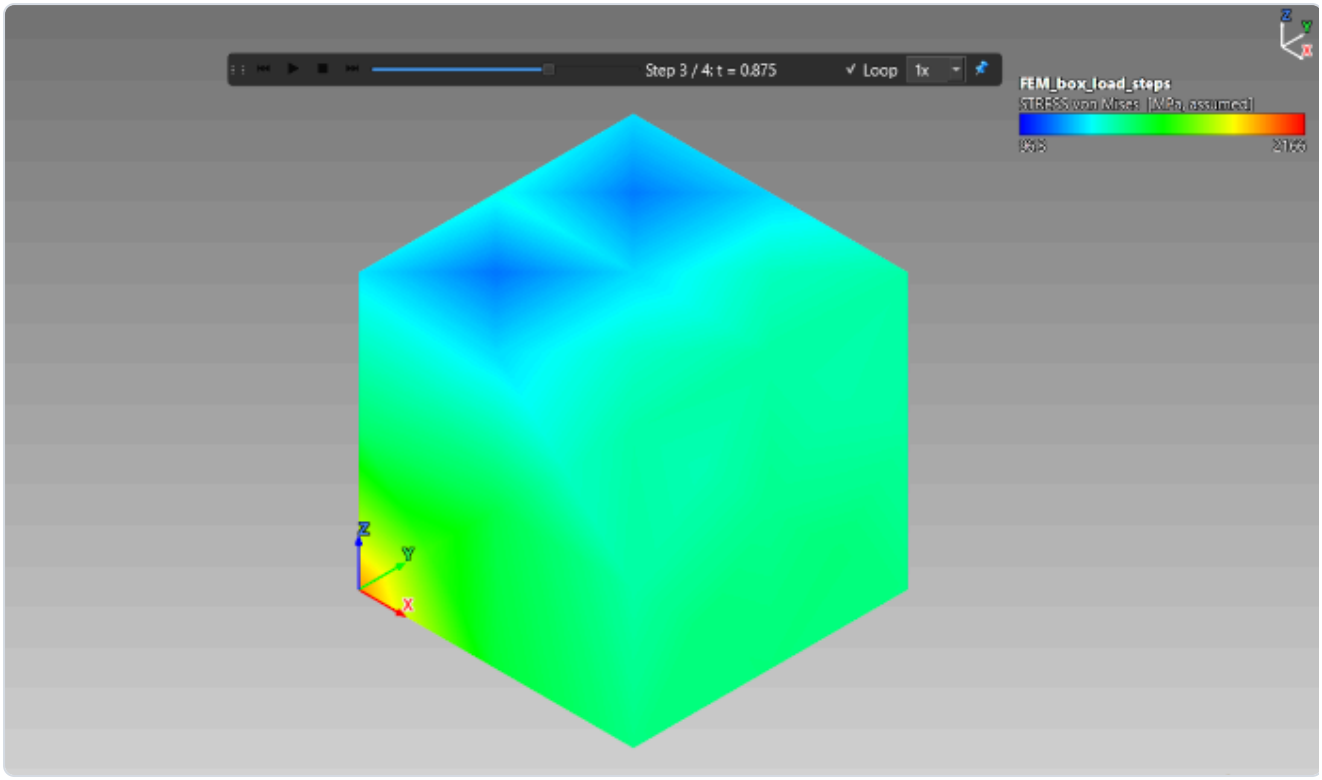
The word **step** covers three different things, and ModelViewer treats them alike:

- **Load steps and increments** of a structural analysis (for example 25 %, 50 %, 87.5 %, 100 % of a load);
- **Time steps** of a transient thermal or flow analysis (an OpenFOAM case has one per time directory);
- **Modes** of a frequency analysis (chapter 4).

A result with more than one step shows a **timeline** at the top of the viewport. A result with a single step has no timeline.

5.2 The timeline

Control	Use
Play / Pause	Plays the steps in order at the chosen speed.
Previous / Next step	Moves by one step.
Stop	Stops and rewinds to the first step.
Slider	Jumps to any step; drag it to scrub through time. The label shows the step number and its time (or the mode's frequency).
Loop	Starts again from the first step at the end.
Speed	Playback speed, from slower to faster than real time.
Pin	The bar can be dragged to another place in the viewport (double-click resets it); the pin locks it where it is.



The timeline at the top of the viewport while playing the load steps of the box

5.3 Colours over time

By default the colour range covers **all steps** (chapter 3), so a colour means the same value in every frame and you can watch a stress build up or heat flow in. Choose **Automatic (this step)** when you want each step to use its own full range - for instance to see the *shape* of a mode - and **Custom** to fix the scale yourself.

5.4 Charts and time

The timeline and the time charts of chapter 7 drive each other: an orange cursor in a chart marks the current step, and clicking or dragging in the chart moves the result to the nearest step. When a result and an animated 3D plot share one playback bar (chapter 18), the bar follows physical time.

5.5 Long transient results

A result with many steps and many cells holds a lot of data. Results whose step data would need more than 256 MB, and which have more than one step, are read **lazily**: only a few steps are in memory at a time and the others are read from the file when they are shown. Opening is fast and memory stays flat, at the price of a short read each time you move to a step that is not resident, and of the first scan for the all-steps colour range. Chapter 9 explains this and its consequences.

Samples

File	Steps
`FEM_box_load_steps.frd`	Four increments of a load ramp; play it to watch the stress grow on a fixed colour scale.
`FEM_box_thermal_transient.frd`	20 time steps over 10 s: one face held at 100, the rest starting at 20.
`FEM_box_modes.frd`	The first six vibration modes, labelled with their frequencies.
`openfoam_cavity/cavity.foam`	Five time directories, 0 to 2 s (synthetic values; the case exercises the reader, the cell-data display and the units).

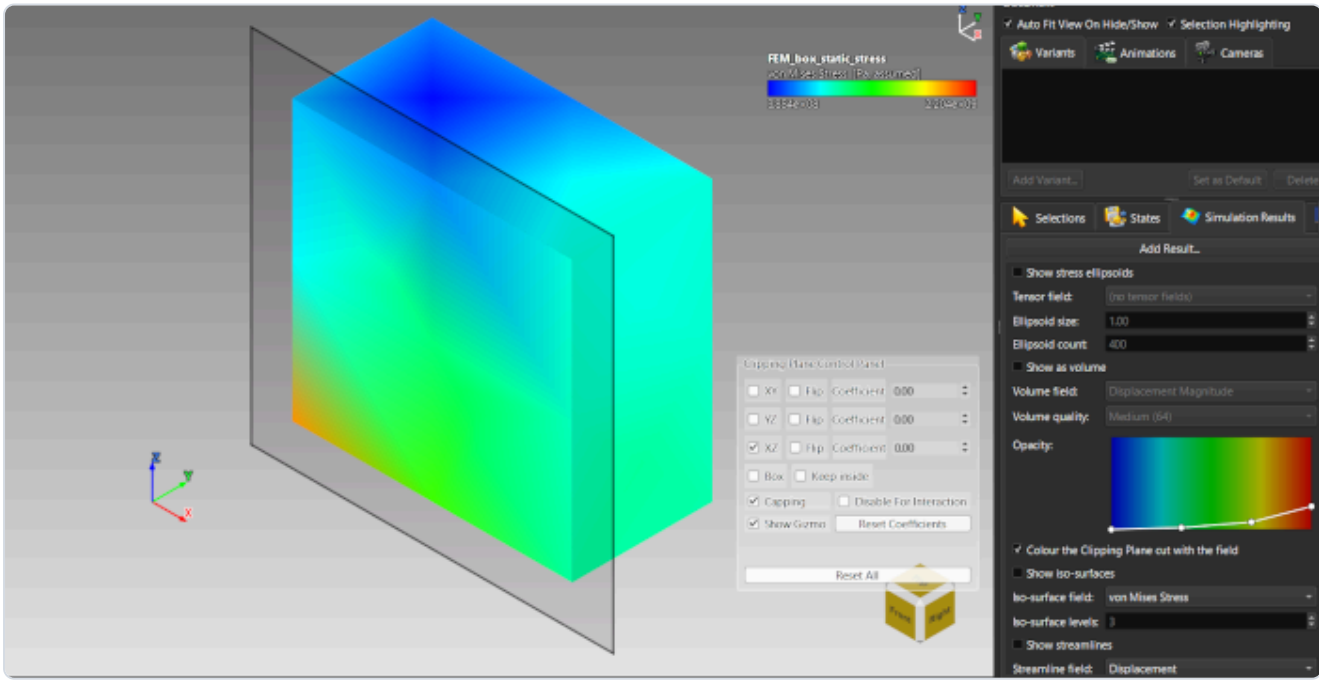
6 Looking inside a result

The surface of a solid shows only half the story: the highest stress is often inside a section, the hottest region under the skin, the fastest flow in the middle of a duct. ModelViewer offers six ways to look in, from the simplest (a cut) to the most specialised (stress tensors as glyphs). The first four need **volume cells** (tetrahedra, hexahedra, wedges, pyramids, polyhedra); on a shell result the panel states that there is nothing to cut or trace. All six are **overlays**: they are drawn with the result but are not scene meshes, so they cannot be selected, exported or drawn by the ray tracer.

6.1 Sections: the cut coloured by the field

The Clipping Planes (the Section View button of the View toolbar; see the tutorial lesson 12) cut a result like any model. Tick **Colour the Clipping Plane cut with the field** and the cut face shows the shown field - with the colours of the legend - instead of an empty hollow. The coloured cut follows the plane as you move it.

- It works with a node field (smooth) and with a cell field (flat per cell).
- With several planes switched on, a plane's cut is kept only where every other plane has removed the material, so the cuts always lie on the boundary of what is left.
- Only the axis-aligned planes take part, not the box mode and not custom planes.
- The cut is computed on the volume cells, and concave polyhedra are supported.

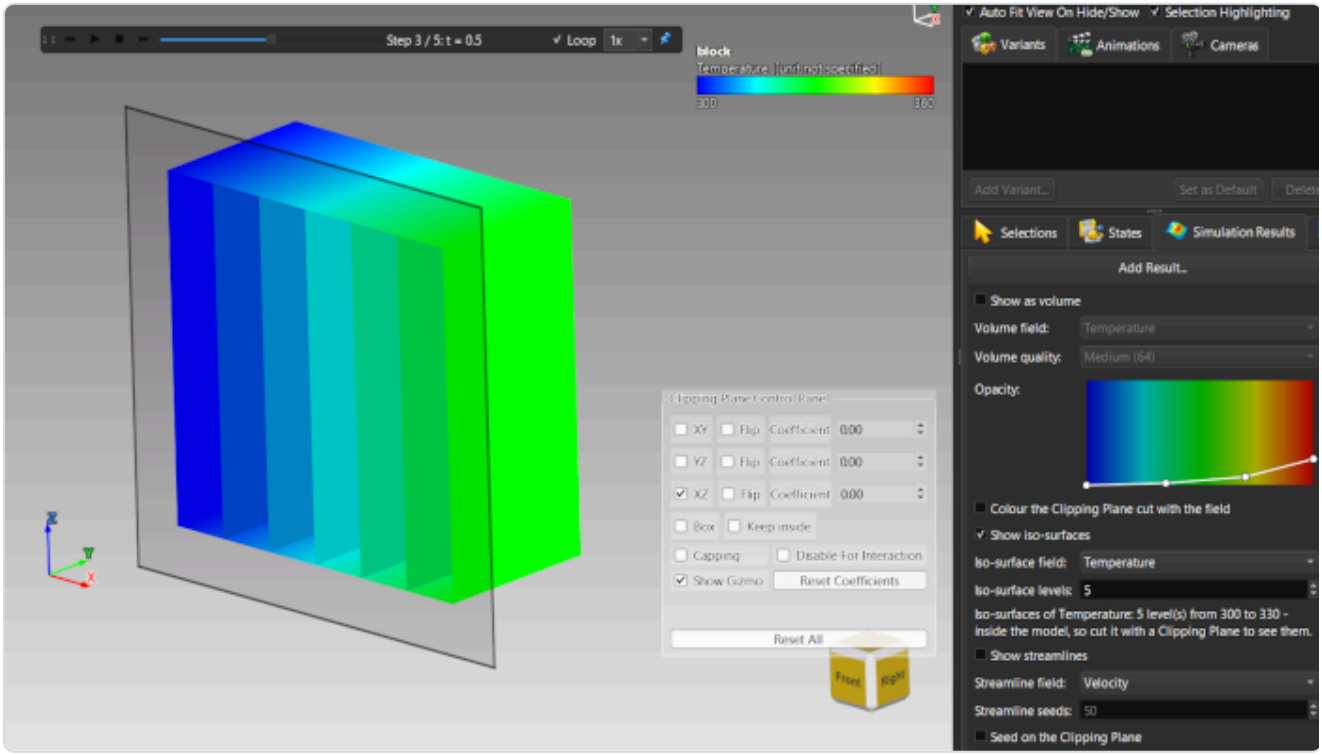


A box cut by a clipping plane, the cut face coloured by stress

6.2 Iso-surfaces

An **iso-surface** is the surface on which a field has one constant value - a shell of equal temperature, a layer of equal stress. **Show iso-surfaces** draws a set of them for the **Iso-surface field** (a node scalar, or the magnitude of a node vector), at **Iso-surface levels** numbering 1 to 20, spaced evenly strictly inside the range of the field at the current step, each in its own colour. A constant field has no iso-surface; the info line says so.

- Iso-surfaces lie inside the model, so they are seen through a cut. Switch on a Clipping Plane: the section cap is kept open while iso-surfaces are shown, because an opaque cap would hide them. The opaque coloured cut does hide what is behind it; switch that option off to see the iso-surfaces.
- They are recomputed when the step or the field changes.



Iso-surfaces of the temperature field inside a block, opened by a clipping plane

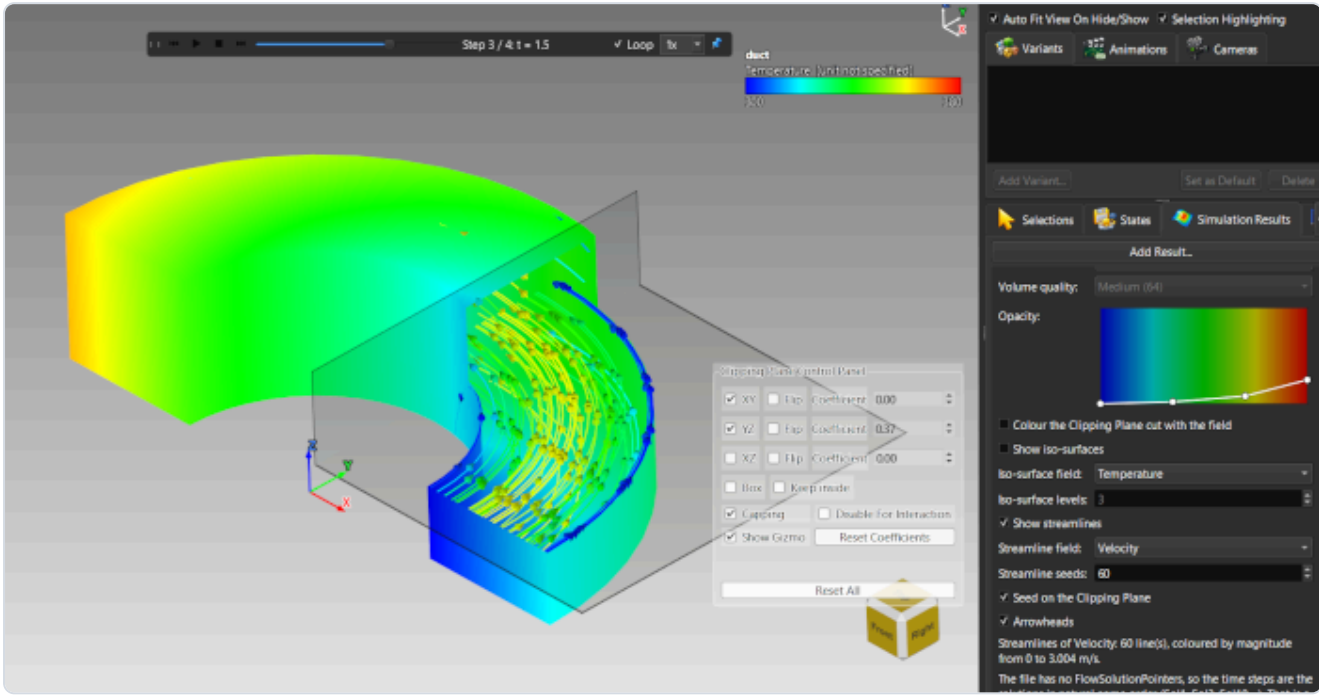
6.3 Streamlines

Streamlines follow a velocity field: each line is the path a massless particle would take if the flow were frozen at the current step. **Show streamlines** uses a node vector field (**Streamline field**); **Streamline seeds** (1 to 500) is the number of start points.

- **Seeding.** By default the seeds are random points of the volume - the same ones every time, so animation frames stay comparable. **Seed on the Clipping Plane** starts the lines from the cut faces of the enabled planes instead, weighted by area, which shows exactly where the flow goes from a cross-section. A seed outside the mesh, or where the field is zero, gives no line.
- **Tracing.** Lines are integrated with a fourth-order Runge-Kutta method in both directions from a seed until they leave the mesh, the speed falls to about a millionth of the largest, or a limit is reached (2000 steps, or four model diagonals, per direction). The field is interpolated exactly inside the cell it passes

through.

- **Drawing.** Lines are coloured by the magnitude of the field, with the legend's range when the surface shows the same field. **Arrowheads** adds small cones that show the direction, and **Line thickness** sets the width. Lines are trimmed to what the planes leave visible.
- **Limits.** Only node vector fields are traced. Concave polyhedra are excluded (the panel reports how many), because the interpolation is only valid in convex cells. Tracing is not threaded, so a very large result may retrace visibly at each time step.

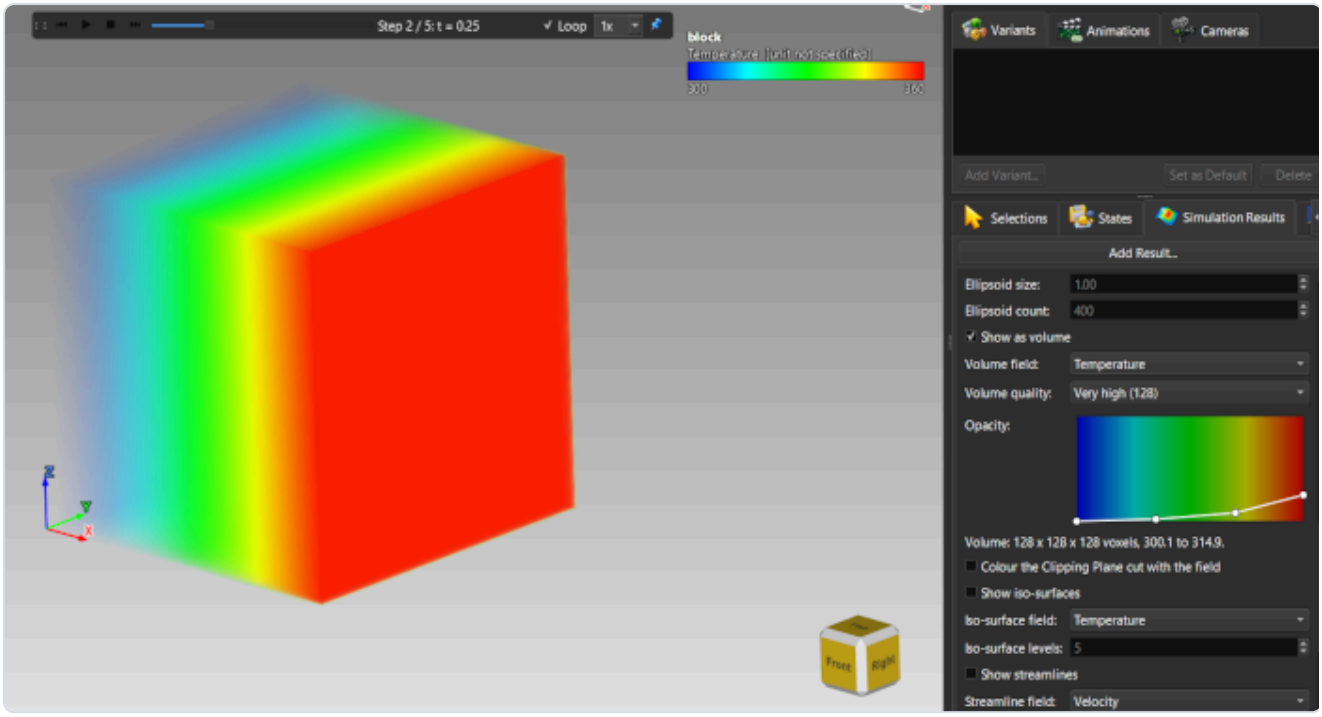


Streamlines through a swirling duct flow, with arrowheads

6.4 Volume rendering

Show as volume renders a field as a semi-transparent cloud instead of a surface, by ray marching through the volume. Choose the **Volume field** and a **Volume quality** (Low 32, Medium 64, High 96, Very high 128 samples along each ray: more is smoother and slower).

The **opacity curve** beneath the options says how opaque each value is: the horizontal axis is the normalised field value, the vertical axis the opacity. **Drag** a point to move it, **double-click** to add a point, **right-click** an inner point to remove it (the two end points stay at 0 and 1). Raising the curve for the hot values and lowering it for the cold ones leaves only the hot core visible; the colour ramp is the one chosen for the surface. A cell field is shown as its averaged values. The volume shows the current step.



The temperature field of a block rendered as a volume, with the opacity curve

6.5 Vector arrows

Show vector arrows draws an arrow for a vector field at sampled points of the surface: choose the **Arrow field**, the **Arrow size** (1 makes the largest arrow 5 % of the model size; the size stays stable while you zoom) and the **Arrow count** (about this many arrows, spread evenly over the surface). With **Scale arrows by magnitude** off, every arrow has the same length and only the colour shows the magnitude - useful when a few large vectors would make the rest invisible.

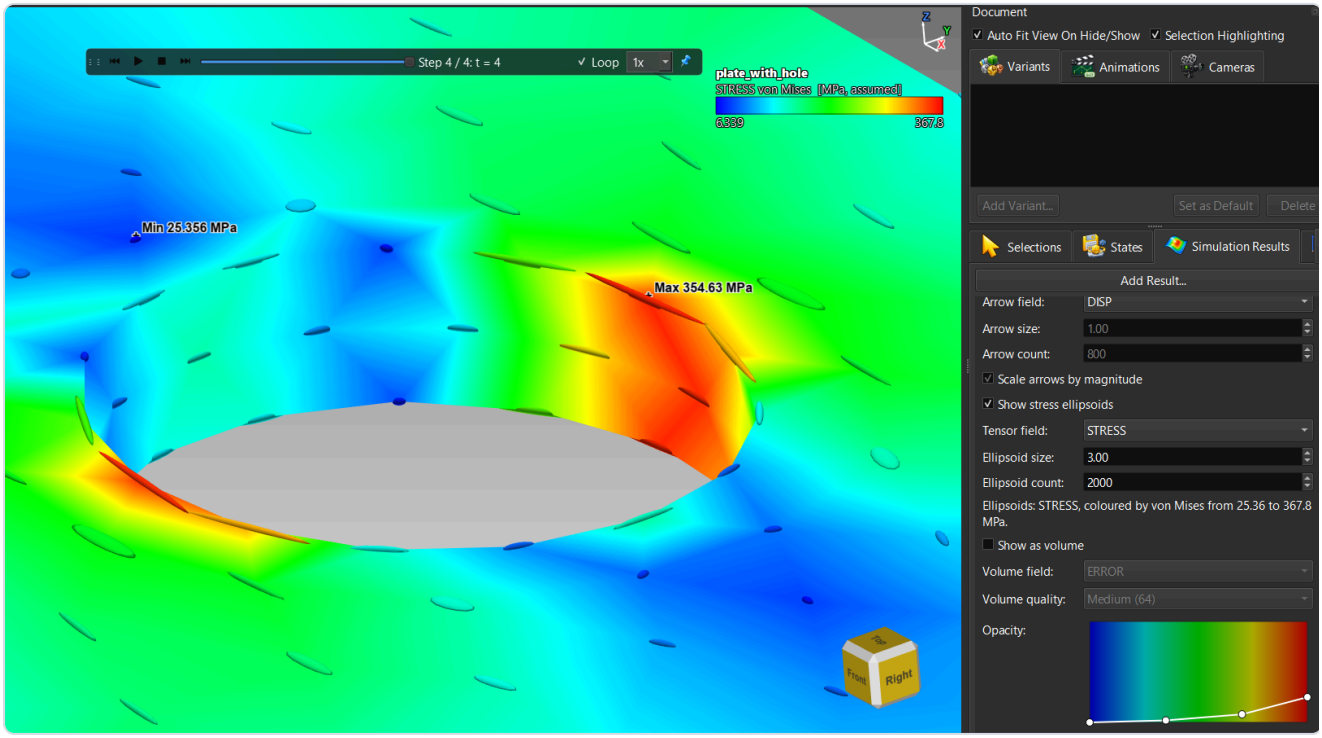
6.6 Stress ellipsoids (tensor glyphs)

For a symmetric stress tensor, **Show stress ellipsoids** draws an ellipsoid at sampled points of the surface. Its three axes point along the **principal directions** and their lengths follow the **principal values**. The longest axis is always the same size on screen, so the *shape* shows how the stress is distributed - a needle for a pull in one direction, a flat disc for a two-way pull, a sphere for an even pressure - and the *colour* shows how large it is.

- **Tensor field**: every six-component field of the result whose name contains *stress*; the node fields come before the cell fields in the list. A tensor with another name (an OpenFOAM *sigma*, say) is not offered, because six components alone do not make a stress.
- **Ellipsoid size** (stable while zooming) and **Ellipsoid count** (20 to 5000). Raise the count to 1500 or more for a plate.
- The choices are saved with the session.

Good first tests: `plate_with_hole.frd` (needles along the pull far from the hole, changing sharply around it; the stress concentration is about three times the far-field stress), `ibeam_cantilever.frd` (bending: needles in the flanges, tilting in the web) and `stress_states.vtu`, a synthetic block with three tensors

made to try the **Tensor field** choice.



Stress ellipsoids around the hole of a plate

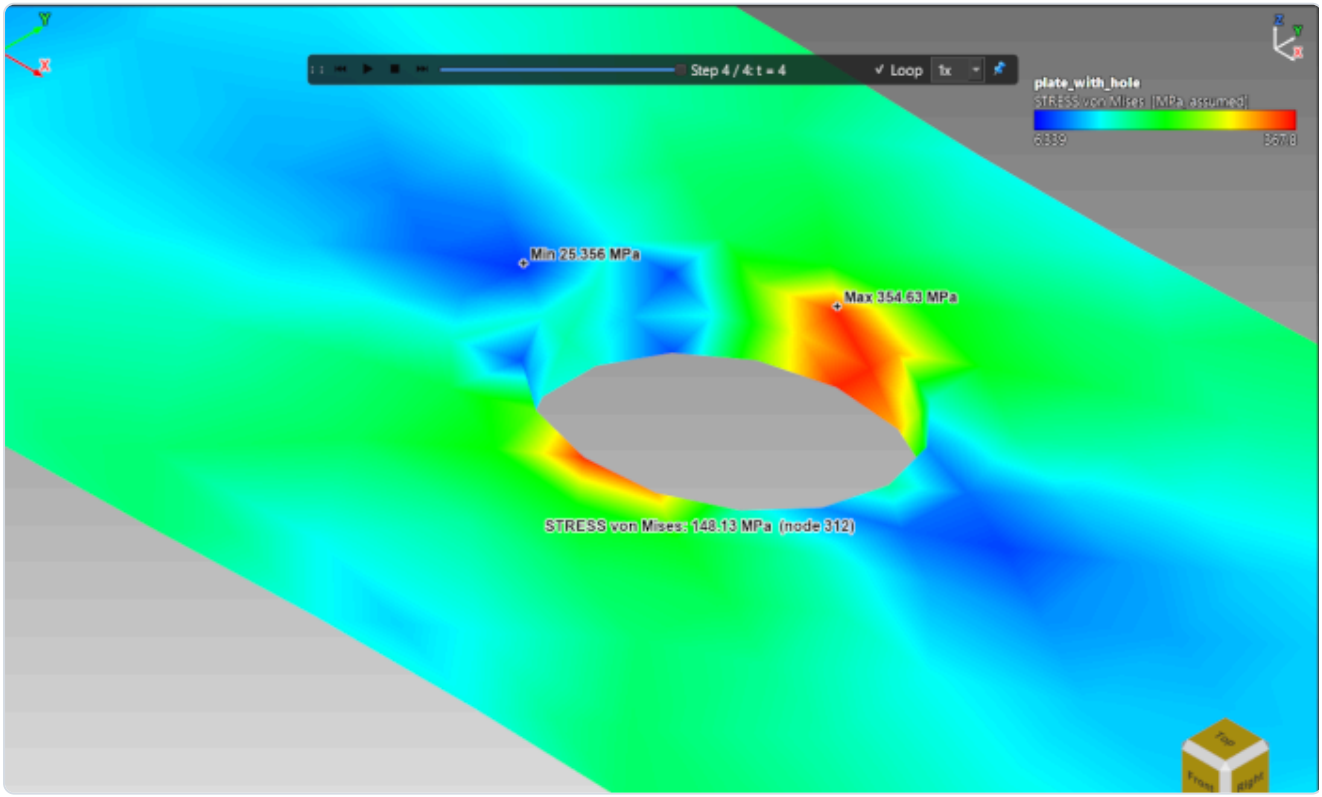
> **TIP**
> The overlays combine: a deformed shape with a coloured section and a few streamlines is a perfectly good picture. Switch off what you do not use, though - the picture is clearest with one idea per view.

7 Probing and charts

Colours tell you *where* a value is high or low; a chart tells you *how much*, and how it changes. ModelViewer reads values straight from the result and plots them without leaving the program.

7.1 Reading a single value

Move the mouse over the model: the value of the shown field at the cursor appears next to it, in the displayed unit. On a node field the value is interpolated at the cursor; on a cell field it is the value of the element under the cursor (or the averaged value, if **Average cell data to nodes** is on).

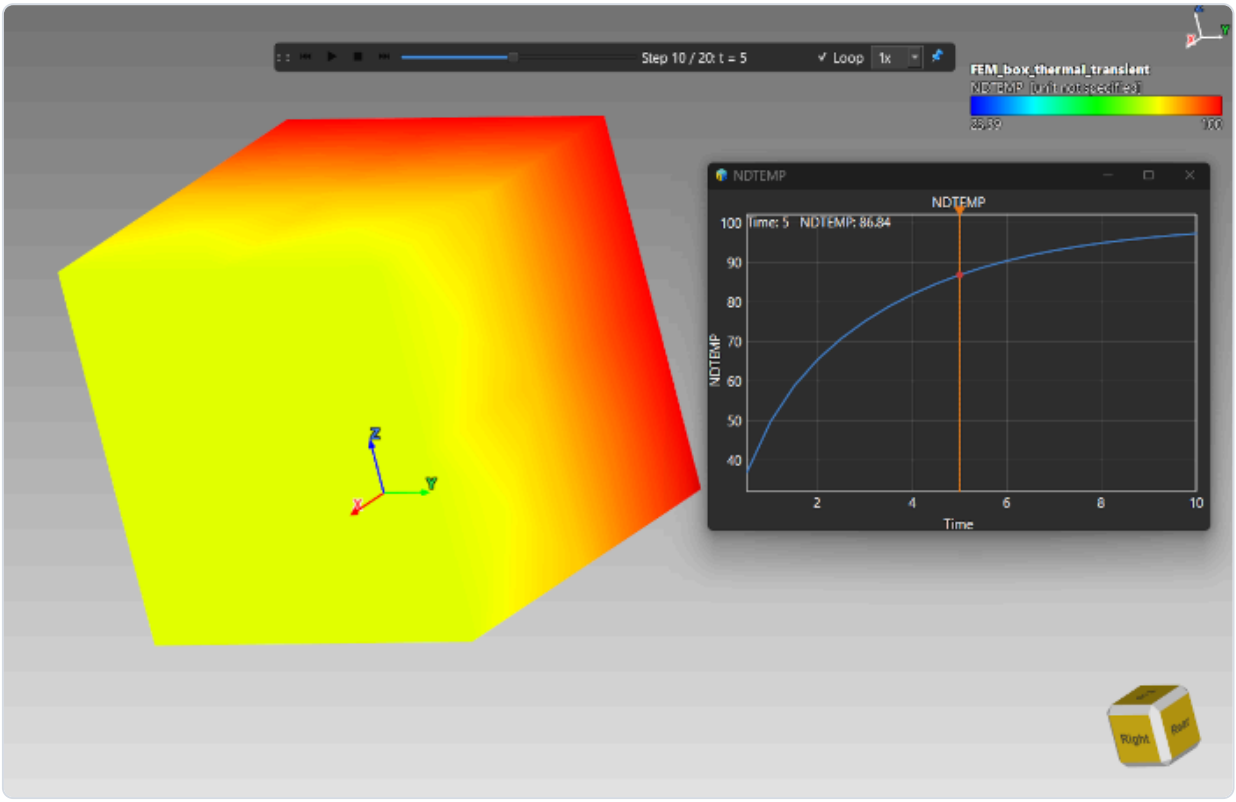


The value under the cursor, read directly from the model

7.2 The four charts

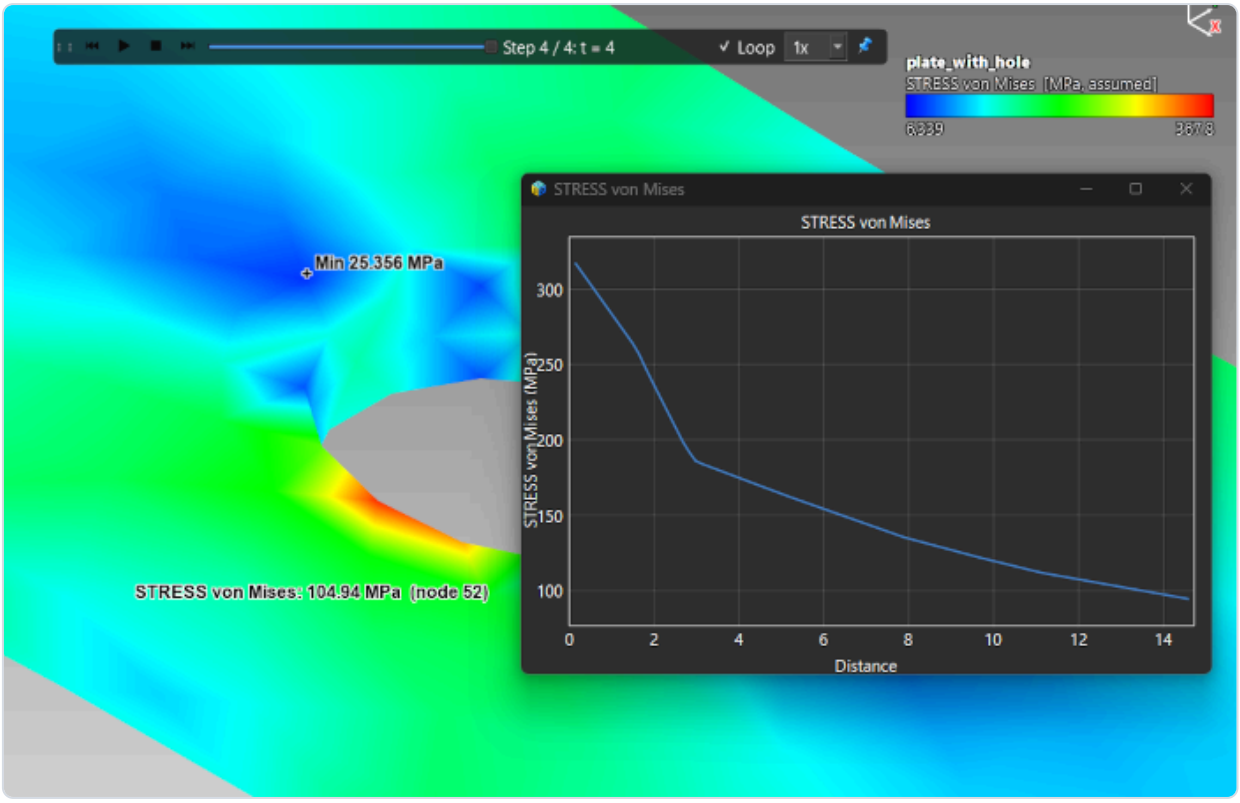
Chart	How to start it	Shows
Plot Over Time	Right-click a point on the result → **Plot Over Time Here**; or the **Plot Over Time…** button, then click a point	The history of the shown field at that point, over all steps. For a modal result the x axis is the frequency of each mode.
Plot Over Line	**Plot Over Line…**, then click two points	The field along the straight line (chord) between the two points, at the current step.

Chart	How to start it	Shows
Distribution	**Distribution...**	A histogram of the field over the whole result: how much of the model is at low, middle and high values.
Added curves	Right-click inside a chart	More points of the model, or your own data from a CSV file, on the same axes.

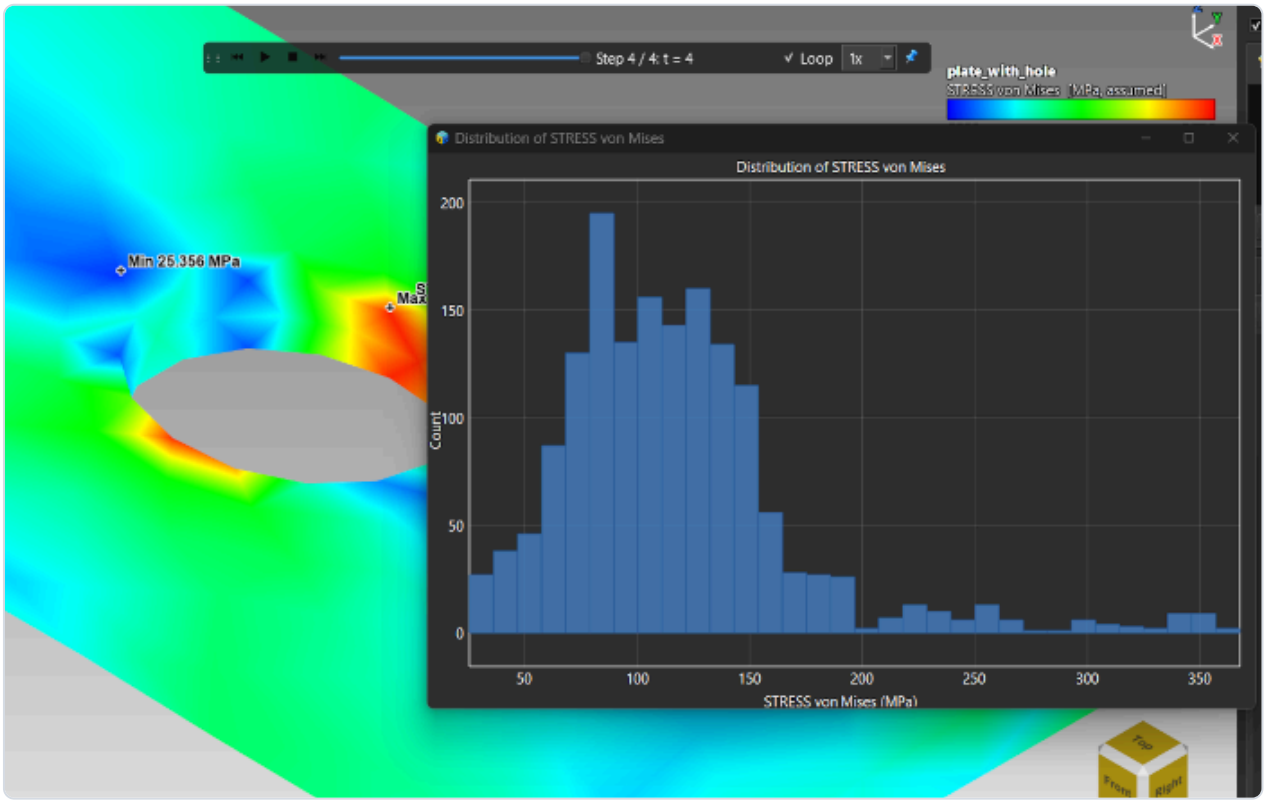


A Plot Over Time chart; the orange cursor marks the current step

- The point history reads the value at the clicked point of the *volume* or, for a shell result, at the closest point of the surface. The chord of a Plot Over Line follows the surface of a shell; where the chord leaves a curved shell the curve has a gap.
- A chart of a cell field uses the averaged values; a result with one step shows its history as a single dot.
- **Plot Over Time Here** is not available while two results are being compared (chapter 8).



Stress along a line from the edge of the hole to the edge of the plate



A histogram of the von Mises field

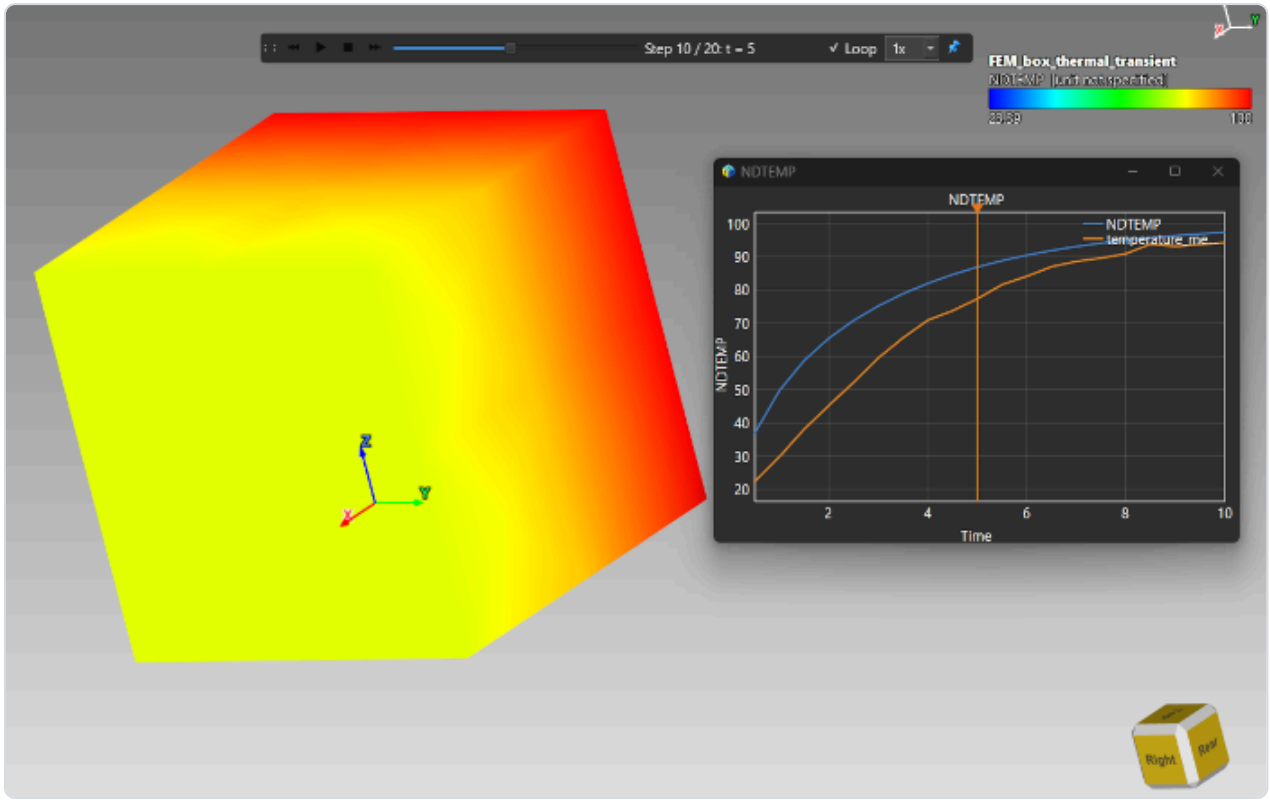
7.3 Working in a chart

Action	Result
Mouse wheel	Zoom around the pointer. With Shift only the x axis zooms, with Ctrl only the y axis.
Drag with the middle button	Pan.
Double-click, or right-click → **Reset zoom**	Show everything again.
Click or drag in a time chart	Move the result to the nearest step (the orange cursor marks the current one); the chart and the timeline drive each other.
Right-click → **Add a point from the model**	Then click another point on the same result: its history joins the chart as another curve, named by its position, with a legend.
Right-click → **Add curve from CSV…**	Adds your own data (see below).
Right-click → **Remove added curves**	Clears the added curves again.
Right-click → **Save image…**	Writes the chart as a PNG image.
Right-click → **Export data (CSV)…**	Writes the curves as a CSV: one x, y pair of columns per curve, with the unit in the header.

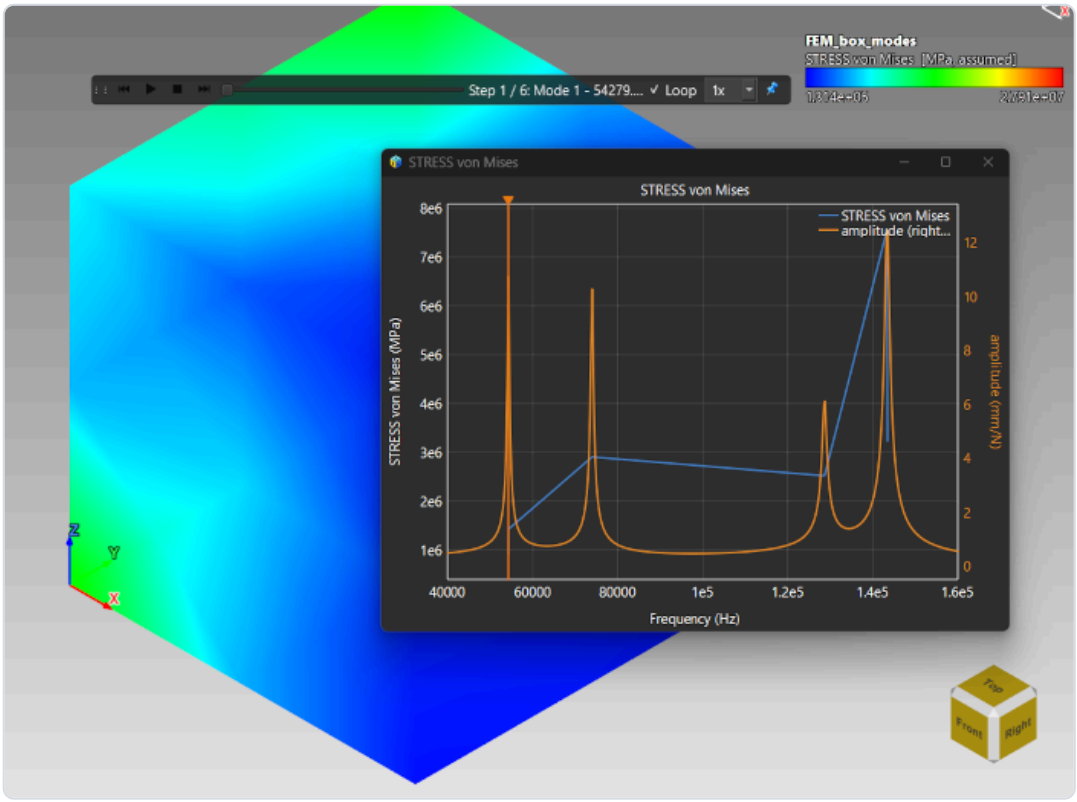
7.4 Adding your own data

A chart can carry measurements, a response curve or another analysis next to the simulated curve. The CSV needs **two columns, x then y**, and a header row that names the curve. A header of the form `Name (unit)` gives the curve a unit: if that unit is the same as the main curve's, both share the y axis; if it differs - a temperature next to a stress - the curve is drawn against a **second y axis on the right**. A curve whose header has no unit shares the main axis.

Sample	Use
`thermal_probe_test.csv`	Synthetic "thermocouple" readings (time, temperature) for the middle of the far face of `FEM_box_thermal_transient.frd`. Open the result, plot over time at that face, add the CSV, play: the cursor travels along both curves. The clicked point is not exactly the sample's, so the curves are close, not identical.
`FEM_box_modes_frf.csv`	A synthetic frequency-response curve with peaks at the six mode frequencies of `FEM_box_modes.frd`. Plot over time on any point, add the CSV, and click a peak: the result jumps to that mode.



The simulated history and the measured readings on one chart



Mode frequencies on the x axis with a response curve and its peaks

TIP

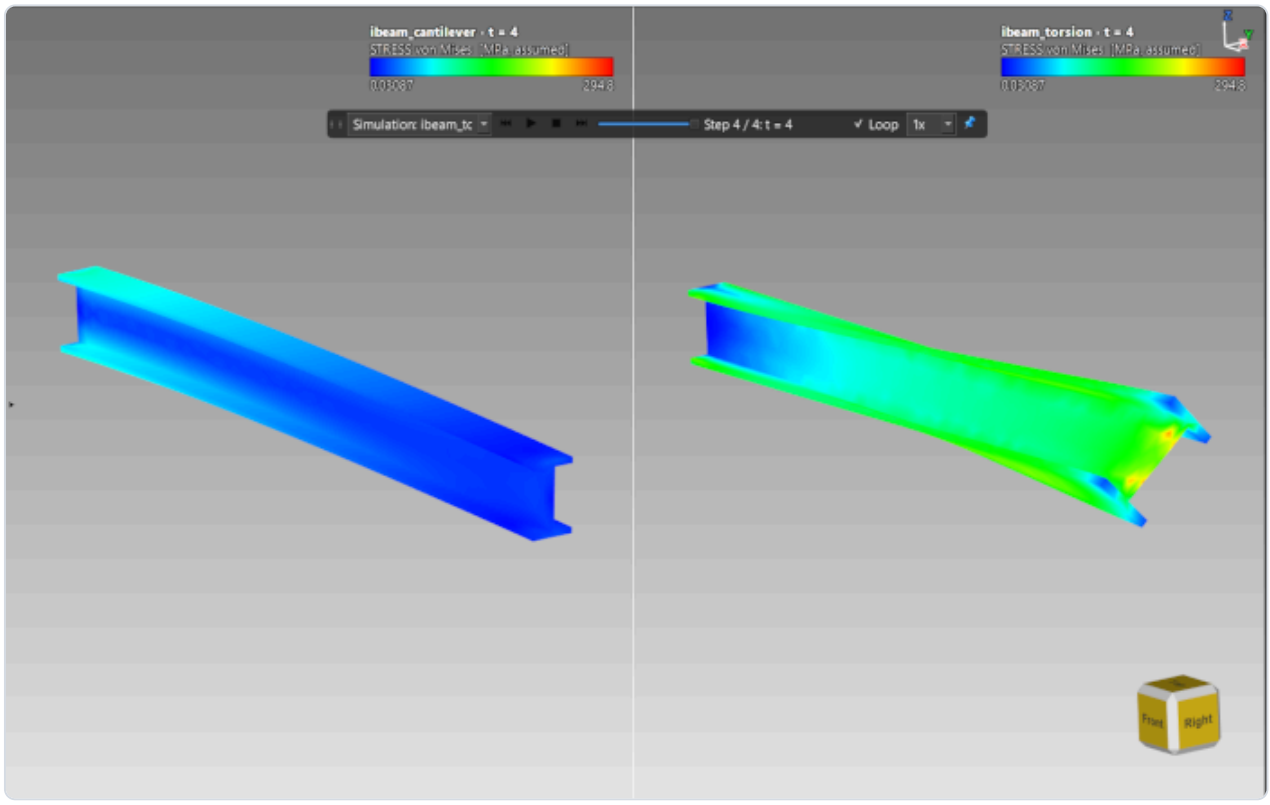
An exported chart CSV can be added back to a chart later with **Add curve from CSV...** - handy for comparing two analysis runs on one chart, because the header carries the unit.

8 Comparing results

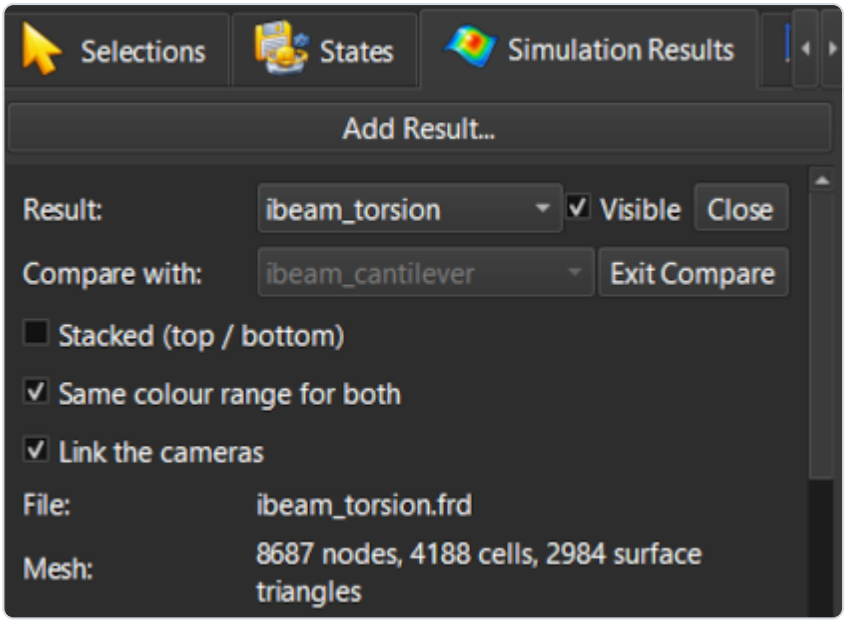
The usual reasons to open two results are two designs, before and after a change, a static run against a vibration mode, or two load cases. Comparing them needs the results side by side, seen from the same viewpoint, with colours that mean the same thing. **Compare** does exactly that.

8.1 Setting up a comparison

1. Put both results in **one document**: open the first with **File** → **Open…** and add the second with **Visualization** → **Simulation Results** → **Add Result to This Document…**. Both keep their true file coordinates, so two results of the same model coincide, which is why compare draws them in separate panes.
2. In the Simulation Results tab choose the other result in **Compare with** and click **Compare** (or use **Visualization** → **Simulation Results** → **Compare Results…**). The viewport splits into two panes, each drawing one result with its own legend. **Show in** chooses side by side or stacked (top and bottom).
3. Tick **Same colour range for both**: equal colours then mean equal values in both panes. Without it each result uses its own range.
4. Tick **Link the cameras** to keep both views in step: rotating, panning or zooming one does the same in the other.
5. Click **Exit Compare** (the same button) to return to the single view.



Bending (left) and torsion (right) of the same I-beam, side by side with one colour range



The comparison controls of the tab

8.2 What works in compare mode, and what does not

- Compare shows two **different** results of the document; the same result cannot be compared with itself. Each pane has its own legend, headed with the result's name.
- Orbit, pan, zoom and the standard views work. **Same colour range for both** joins the ranges of the two results, and is only available while both show the same unit.
- The timeline drives the active result only, the one chosen in the **Result** box.
- While compare is on, selection by clicking, the hover highlight and the hover value readout, and the clipping-plane gizmos are switched off, and **Plot Over Time Here** is not available. The measurement and annotation tools are not pane-aware yet. Ray tracing is off.
- **Fit to screen** still fits the whole window, so a wide model can be wider than its pane: zoom out a little.
- Compare ends by itself when either result is hidden, closed or removed (also by undo). The layout is not saved in a session.
- Compare holds both results in memory at once, so two very large results need twice the memory (chapter 9).

> **CAUTION**
> Compare shows two results next to each other; it does not compute the difference between them (there is no A minus B field). To read numbers out of two analyses, chart the same point of each and put both on one chart (chapter 7), or compare the hover values one result at a time.

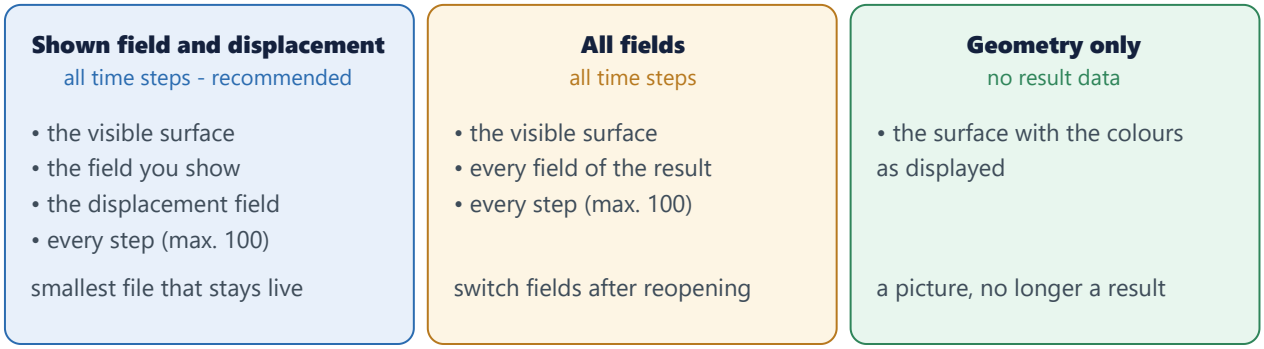
A good pair to try

`ibeam_cantilever.frd` (bending) and `ibeam_torsion.frd` (torsion) are the same I-beam under different loads: add one to the other's document, turn on the deformed shape and the shared colour range, and set the last load step.

9 Saving, large results and limits

9.1 Saving a result in a session

A result is a viewing session like any other, so it is saved in the `.mvf` session file (**File** → **Save**). The point of saving is that the session reopens **without the original solver file**, with the field, range, colormap, units, deformation and timeline as you left them. When the document contains results, ModelViewer asks what to store, once per session:



The three choices of the Save Simulation Results dialog; each shows an estimate of the file size

What is stored

- The snapshot is the **boundary surface**, not the volume: values are stored per surface vertex (not per solver node), compressed without loss, as 32-bit numbers. The dialog shows a size estimate before anything is written.
- The steps (with their times or labels), the fields with their units and quantities, the all-solver-nodes range of each field and step (so the legend and the all-steps colour range are identical to the live result even though interior nodes are not stored), and the view state: field, component, range, colormap, bands, step, deformation and markers.
- The derived stress fields are **not** stored: they are recomputed from the stress tensor on loading.
- At most **100 time steps** are stored per result, evenly spaced; the dialog tells you if that applies.
- The path of the source file is kept as information only; the session never needs it.

Sections, iso-surfaces and streamlines in a saved session

Without the volume, only the visible surface is stored, so the cut faces, iso-surfaces and streamlines on display are saved **as they were** and come back *frozen*: they no longer follow a moved clipping plane or another time step, and the panel says so. Tick **Also store the volume** to keep them live: the nodes, cells and chosen fields of the whole volume are stored too, the restored result is the full result, and everything - sections, iso-surfaces, streamlines, volume rendering - recomputes as before. The price is a much larger file, roughly in proportion to the volume; the dialog shows what the volume adds.

Reopening

- An older version of ModelViewer opens such a session as a plain coloured mesh (the shown colours are stored with it) and ignores the rest.

- If you edit the mesh of a stored result after loading - split, merge, repair - and save, the vertex count no longer matches the stored data; on the next load the result session is dropped with a message rather than showing wrong values. Moving, rotating or scaling the model is harmless.

9.2 Large results

Where the memory goes

Geometry is cheap; **time steps** are what grow. A result holds, for every field and every step, four bytes per value:

Mesh (hexahedra)	Geometry	One step of a node scalar, a node vector and a cell scalar	50 steps, all read at once
1 million cells	about 50 MB	about 20 MB	about 1 GB
5 million cells	about 250 MB	about 100 MB	about 5 GB
20 million cells	about 1 GB	about 410 MB	about 20 GB

These are sizes of what is held in memory; Compare mode holds two results at once.

Lazy step loading

To keep large transient results usable, steps are loaded **on demand**. A result is read lazily when the data of its steps would need more than **256 MB** and it has more than one step; the formats that support it are **Exodus**, **CalculiX FRD**, **CGNS**, **VTKHDF** (unstructured grids and poly data) and **OpenFOAM** cases. The other formats, and VTKHDF image data, are read in full; MED is not lazy yet.

- Opening reads only the mesh and the field headers; the last four steps that were shown stay in memory, the least recently used going first. An OpenFOAM case reads one time directory per step. A CalculiX file is memory-mapped and indexed once.
- A step that is not resident is read from the file when you move to it, so scrubbing a lazy result is slower than a resident one, and **the file stays open** (a CalculiX `.frd` is locked for writing while the result is loaded).
- The range over all steps - the default colour range - reads every step once on first use; a busy cursor covers that first scan.
- **Saving** a lazy result reads the whole file, because the snapshot stores every step it keeps.

Measured on a result of one million hexahedra with five steps and nine fields (485 MB on disk), loaded fully and lazily:

	Fully read	Lazy
Dataset in memory after opening	361 MB	47 MB
Time to read and build	1.0 s	0.15 s
Process memory peak	584 MB	323 MB
Show one step	3 ms	0.18 s (read from the file)

	Fully read	Lazy
Range over all five steps	7 ms	0.9 s

Figures from a Windows release build; the colour ranges agree to the last digit between the two modes. Your numbers depend on the disk and the format - binary HDF5 and NetCDF files read much faster than ASCII ones.

Practical advice

- Write the fields and steps you need from the solver; every extra step costs time and disk space, in the file and in a saved session.
- Prefer binary, structured formats (VTKHDF, Exodus, CGNS) over ASCII `.vtk` or `.frd` for big transient results.
- Opening takes time in two stages: reading the file, then extracting the boundary surface (a few seconds for five million cells; the extraction is done in parallel).
- Close results you no longer look at, and avoid Compare with two huge results.

9.3 Limits and what is not supported

Limit	Details
No solving or editing	ModelViewer neither meshes nor solves, edits values, or creates difference (A minus B) fields.
Open formats only	No proprietary formats (Abaqus `.odb`, ANSYS `.rst`, Nastran OP2); export from those programs to VTK or another open format first.
Shells have no inside	Sections, iso-surfaces, streamlines and volume rendering need volume cells.
Overlays are not meshes	Cut faces, iso-surfaces, streamlines, arrows and ellipsoids cannot be selected, exported to glTF/OBJ, or drawn by the ray tracer. Results themselves are drawn flat in every rendering mode and are not ray traced.
Streamline fields	Node vector fields only; concave polyhedra are excluded from tracing.
Section tools	The coloured cut and the seeding on a cut use the axis-aligned Clipping Planes, not the box mode or custom planes.
File-format gaps	Composite VTKHDF files (multi-block), structured and rectilinear VTKHDF grids, and CGNS face-centred data are not read yet; Exodus beam networks load but have nothing to draw; legacy VTK polygons, strips and vertices are not displayed.
Compare	Two results at a time, not saved, and not pane-aware for picking, measurement and annotation (chapter 8).

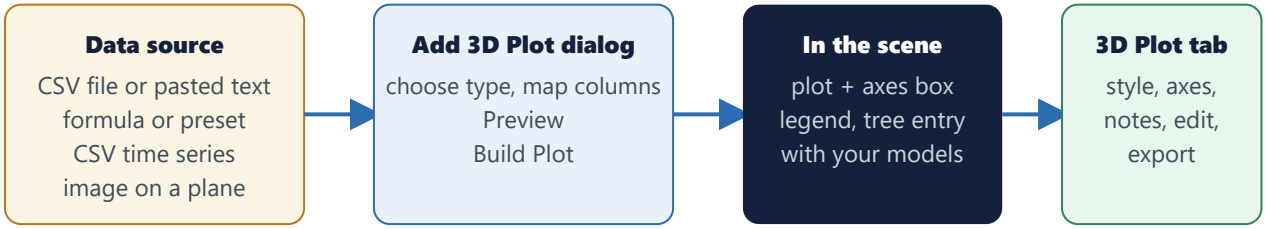
PART II

3D plotting

10 Introduction to 3D plotting

ModelViewer can plot data that has nothing to do with a solver. A table of numbers - measurements, an export from a spreadsheet, a mathematical function - becomes a surface, a curve, a cloud of points, a set of bars, a volume or a field of arrows, **in the same 3D scene** as your models, with the same camera, selection, section, annotation and image-export tools, and the same lighting. It is a way to look at data in space, not a spreadsheet or a statistics package: there is no data editing, no fitting, and no live data feed.

10.1 The idea in one picture



From a table of numbers to a plot in the scene

10.2 Data sources and plot types

A plot has two independent choices: **where its data comes from** (the data source) and **what it looks like** (the plot type).

Data source	Gives	Chapter
CSV file or pasted data	A table you map to X, Y, Z and more, drawn as any plot type	11, 12
Formula surface	z(x, y) from an expression or one of 11 presets	13
Parametric surface, parametric curve	x, y, z as functions of two parameters (u, v) or one (t); Torus, Klein bottle, helix, trefoil knot ...	13
Implicit surface	The surface where f(x, y, z) = 0: sphere, torus, gyroid ...	13
Formula vector field, formula streamlines	Arrows or flow lines of a field given by three expressions	13
Formula pathlines, CSV time series (pathlines)	Particle tracks through a field that changes with time	14
Image on a plane	A picture standing on one of the plot planes	16

Plot type	For data that is …
Surface	a height z over x and y (a grid or scattered points)
Contour	the same, drawn as iso-lines of z
Line / Curve	an ordered path of points
Scatter	unordered points (optionally with stems, error bars or fills)
Bar / Histogram	heights over a grid, or counts in bins
Voxel / Volumetric	an occupancy or density on a 3-D grid
Vector (Quiver)	a vector at each point

10.3 What you get in the scene

- Every plot is an entry in the **scene tree** with its own visibility checkbox, so it can be hidden or shown like a model.
- An **axes box** frames the plots, with labels, tick marks and numbers; reference planes (XY, XZ, YZ) are available. **All plots of a document share one set of axes** (chapter 15).
- A **colour legend** appears for plots coloured by a value.
- A plot can be edited later: change the data, columns or formulas and rebuild it in place (chapter 17).
- Plots are saved in the `.mvf` session with their data, and point and line plots also export to glTF, GLB and OBJ (chapter 17).

> **NOTE**
> Plots are analysis views. They are drawn the same way in every rendering mode and are not drawn by the ray tracer. If you want a rendered picture of a plot, use the normal viewer's image export.

11 Plotting a CSV file

11.1 What the table must look like

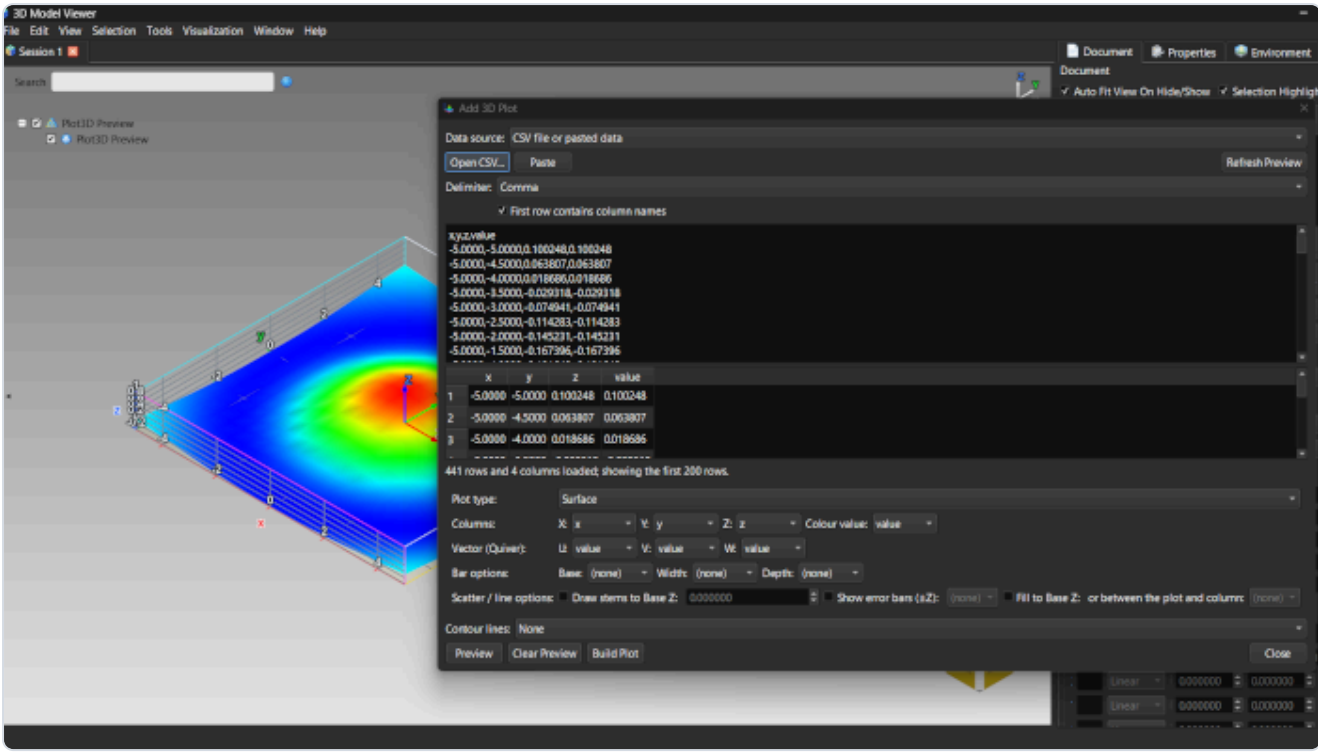
The input is a plain-text table: rows of numbers separated by a delimiter, usually with a **header row** that names the columns. Any spreadsheet or analysis tool can write it. The dialog supports the comma, the semicolon and the tab as the **Delimiter**, and **First row contains column names** says whether the first line is a header. A small example, the start of `surface_ripple.csv` (z = sin(r)/r over a 21 x 21 grid; the last column repeats z as the colour value):

```
x,y,z,value
-5.0000,-5.0000,0.100248,0.100248
-5.0000,-4.5000,0.063807,0.063807
-5.0000,-4.0000,0.018686,0.018686
...
```

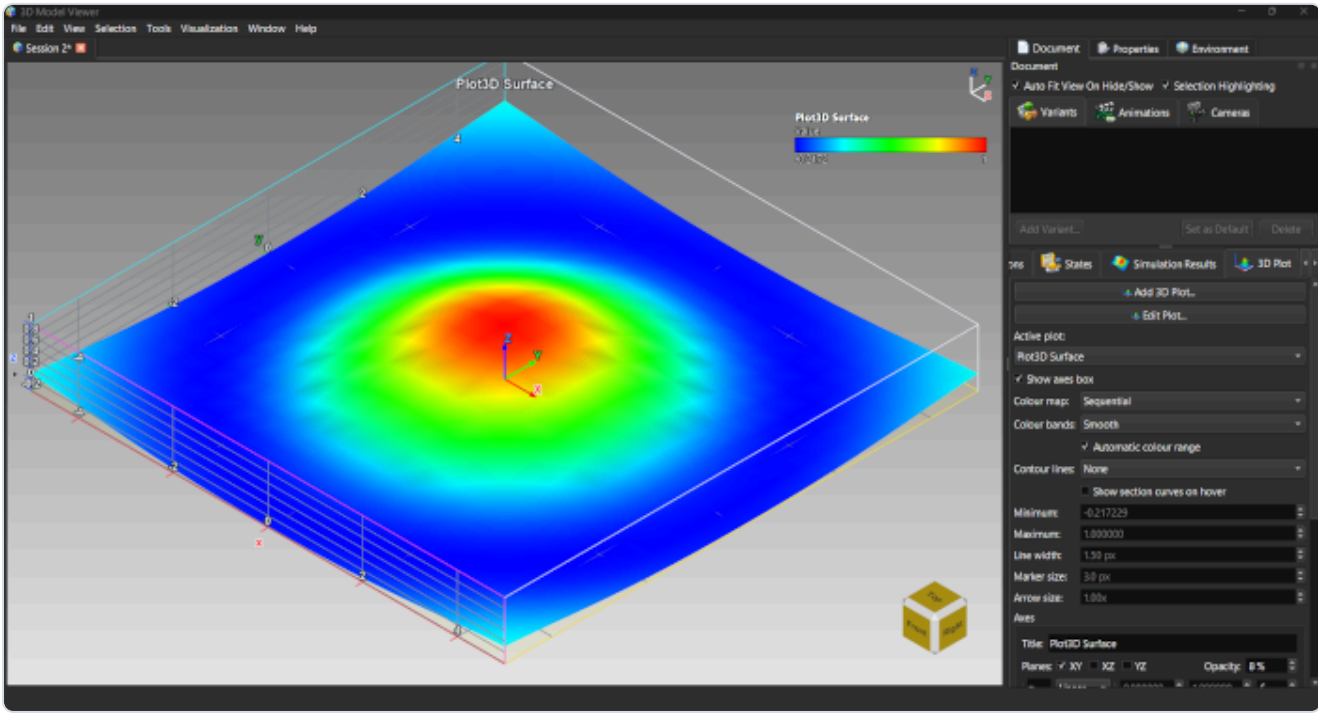
- Numbers use the decimal point. Empty or non-numeric cells in a numeric column make the table unusable; the status line names the row.
- For a **line** the row order matters (it is the order along the path); for the other types it does not.
- The table can be opened from a file (**Open CSV...**) or pasted (**Paste**); after loading, the first 200 rows are shown in a preview table, and the status line reports how many rows and columns were read.

11.2 The Add 3D Plot dialog

1. Open **Visualization** → **Plot 3D...** (or **Add 3D Plot...** in the **3D Plot** tab).
2. **Data source**: *CSV file or pasted data*. Open or paste your table; check the delimiter and the header setting.
3. **Plot type**: pick the type (chapter 12). The column choices below change to match it.
4. Map the columns: the position columns **X**, **Y**, **Z**, and, as the type needs, **Colour value**, **U**, **V**, **W**, **Base**, **Width**, **Depth**. A choice of *(none)* leaves a mapping out. The program guesses sensible columns from the header names, and you can change them: the mapping is by position in the table, so the order of columns in the file is a convenience, not a requirement.
5. **Preview** shows the plot in the scene before it is added - the same size, colours and behaviour as the final plot - and **Clear Preview** removes it. Closing the dialog also removes a preview.
6. **Build Plot** adds the plot to the scene. A **Title** names it.



The Add 3D Plot dialog with `surface_ripple.csv` loaded and the columns mapped



The ripple surface in the scene with its axes box and legend

11.3 The colour value

Surface, line, scatter and bar plots are coloured by a value. By default it is the height (Z); map **Colour value** to another column to colour by something else - the cluster number of a scatter plot, the speed along a path. A scatter plot's colour is per point; a surface's is interpolated between its vertices.

11.4 When the table cannot be used

The status line below the preview says what was read and, if the table cannot be used, why: a row with text where a number is needed (it names the row), a table that is not a complete regular grid where one is required (it says how many rows are missing), or a column that was not chosen. The **Build Plot** button stays disabled until the problem is fixed. Appendix C lists the messages.

> **TIP**
> Start with a sample of the type you want (Appendix A) and look at its columns, then shape your own table the same way. A new file is as easy as a header line and rows of numbers.

12 The plot types

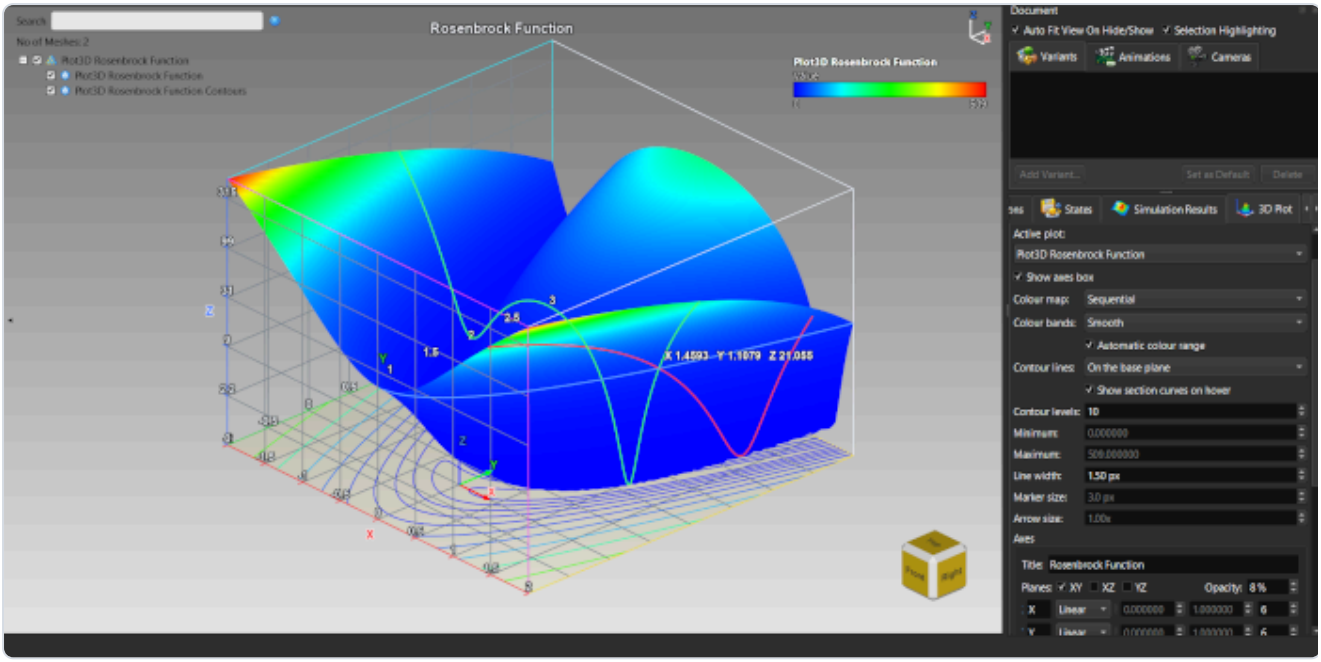
This chapter is the reference for what each plot type needs and offers. The columns are mapped in the Add 3D Plot dialog; the styling controls named here are in the **3D Plot** tab (chapter 15).

Plot type	Columns	Notes
Surface	**X**, **Y**, **Z**; optional **Colour value**	A complete regular grid is drawn as a mesh; scattered points are triangulated (Delaunay). Optionally with contour lines.
Contour	as Surface	Only the iso-lines of Z; they can be flattened onto the base plane.
Line / Curve	**X**, **Y**, **Z**; optional **Colour value**	Points in order. A constant-width line; the width is in screen pixels, so it does not change with the zoom. Can be filled.
Scatter	**X**, **Y**, **Z**; optional **Colour value**	One marker per point. Optional stems, error bars and fills.
Bar / Histogram	**X**, **Y**, **Z** (the height); optional **Base**, **Width**, **Depth**, **Colour value**	One box per row. A one-dimensional histogram sets Y to *(none)*.
Voxel / Volumetric	**X**, **Y**, **Z** (grid indices); **Colour value** (occupancy)	Drawn as a semi-transparent cloud, like the volume rendering of a result.
Vector (Quiver)	**X**, **Y**, **Z** and **U**, **V**, **W**; optional **Colour value**	One arrow per row; arrow size and colour are controlled in the tab.

12.1 Surface and contour

A surface needs z over x and y. When the rows form a **complete regular grid** (every combination of the x values and the y values exactly once, in any row order), the grid is meshed directly. Otherwise the points are triangulated, which is what `surface_scattered.csv` tests: fourteen irregular points with a gently sloped height.

- **Contour lines** adds iso-lines of Z to any surface: *On the surface* draws dark lines on it (visible from both sides), *On the base plane* draws colour-mapped lines flattened onto the floor of the axes box. **Contour levels** sets how many.
- The separate **Contour (surface iso-lines)** type shows the lines only, and **Project contours onto the base plane** flattens them.
- **Colour bands** turns the smooth colour of the surface into discrete bands; a fixed colour range is available too.



Contour lines on the base plane, and the section curves through the hovered point (chapter 16)

12.2 Line / curve

Row order is the order along the curve (`line_helix.csv` : two turns of a helix). The colour follows **Colour value** (or Z) along the curve. The **Line width** is in screen pixels. A line can be **filled** to the base plane or to a second curve (chapter 16).

12.3 Scatter

One marker per row; **Marker size** sets the size. Colours are **per point**, not interpolated: `scatter_clusters.csv` has three clusters coloured by their id. Variants:

- **Draw stems to Base Z**: a thin line from each point down (or up) to the base value.
- **Show error bars (±Z)** with an error column: a vertical bar of ± that size at each point (`scatter_error_bars.csv`).
- **Fill to Base Z**: a translucent ribbon from each point to the base plane, or, with a column chosen in *or between the plot and column*, to a second set of Z values (`scatter_band.csv`). The axes box grows to include both.

12.4 Bar and histogram

- The height is **Z**. **Base**, **Width** and **Depth** are optional; without them bars start at 0 and use a width and depth of 0.8, which leaves a gap between neighbouring bars. A bar below its base extends downward, so signed data works (`bar_signed.csv`).
- **Colour value** defaults to the height.
- **One-dimensional histogram** (`bar_histogram_1d.csv` : columns *bin* and *count*): set X to the bin column, Y to *(none)*, Z to the count, and leave the rest *(none)*. The result is one row of equal bars centred on Y = 0, with height and colour both driven by the count.

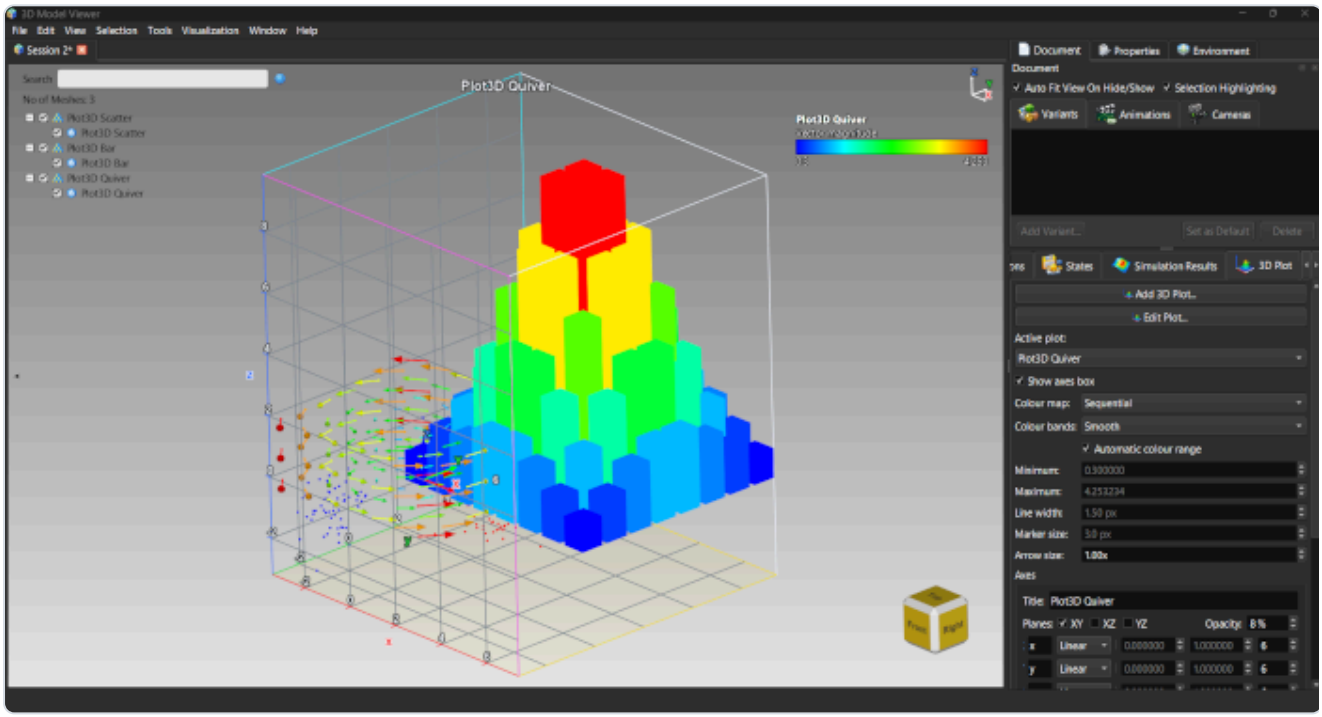
- **Bar width** and **Bar depth** can be changed later in the tab.

12.5 Voxel / volumetric

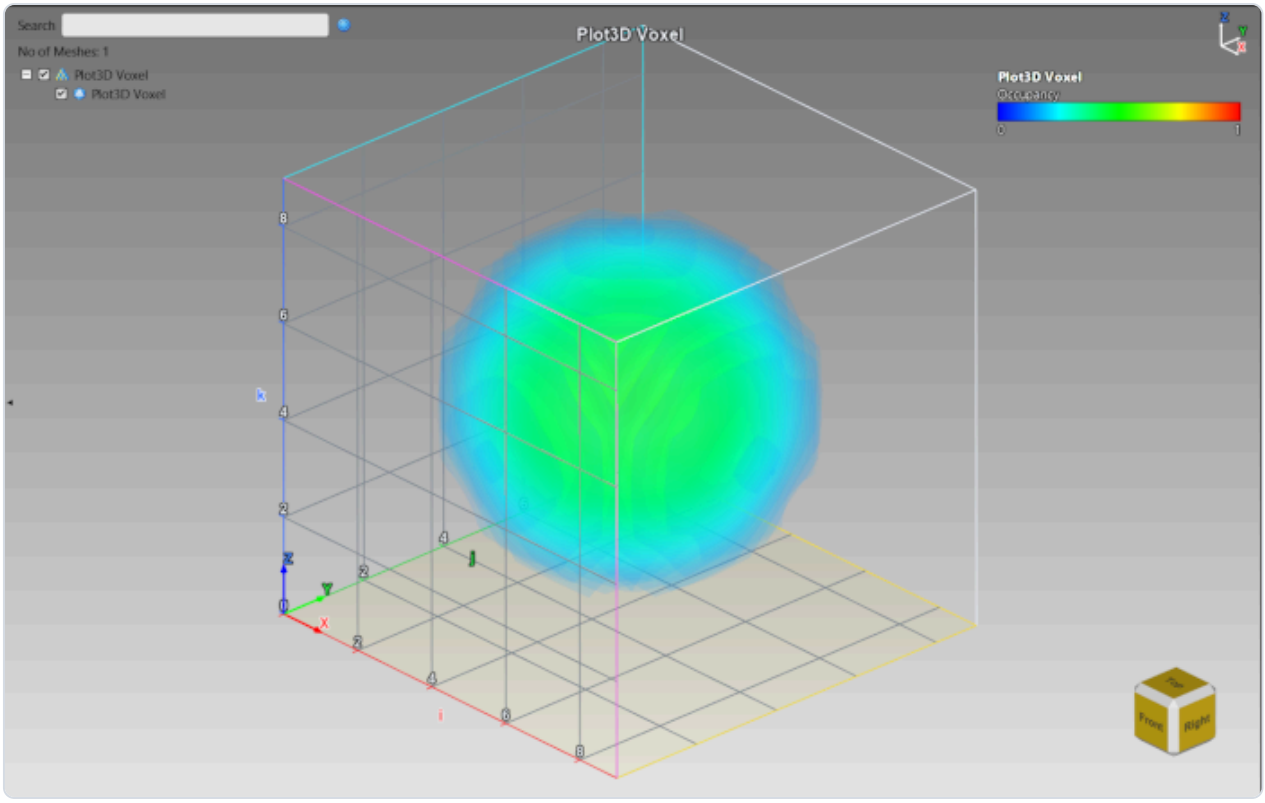
Map X, Y and Z to the grid indices and the colour value to the occupancy (`voxel_sphere.csv` : a sphere carved from a 9 x 9 x 9 grid, with occupancy fading from 1.0 at the centre to 0.15 near the shell). Zero occupancy is transparent and non-zero values fade in by occupancy, so the plot is a soft cloud rather than a set of boxes.

12.6 Vector (quiver)

Each row is a point and a vector. `quiver_vortex.csv` is a rigid rotation about the Z axis plus a small upward component: arrows curl around Z and lengthen with distance. **Arrow size** in the tab sets the length scale without growing with the camera; arrow colour follows the value (the legend reads *Vector magnitude*).



A scatter, a bar chart and a vector field built into one scene



`voxel_sphere.csv` *drawn as a semi-transparent sphere*

13 Plots from formulas

You do not need a file to make a plot: a formula, a parametric definition or a preset generates the data. The sources share the CSV plots' preview, colour, axes, notes and export tools, and a generated plot remembers its definition so it can be edited and rebuilt (chapter 17).

13.1 The sources

Source	You give	Presets
Formula surface	z(x, y) = ..., ranges and sample counts of x and y, named parameters	Plane, Saddle, Paraboloid, Cone, Gaussian, Sinc Ripple, Wave, Mexican Hat, Bivariate Normal, Logistic Regression, Rosenbrock
Parametric surface	x(u, v), y(u, v), z(u, v), ranges of u and v	Torus, Ellipsoid, Mobius Strip, Klein Bottle, Superellipsoid, Helicoid, Catenoid, Enneper Surface, Spherical Harmonic
Parametric curve	x(t), y(t), z(t), range of t	Helix, Lissajous, Trefoil Knot, Viviani Curve, Damped Spiral
Implicit surface	f(x, y, z), drawn where f = 0; ranges and samples of all three axes	Sphere, Torus, Gyroid, Wave Interference
Formula vector field	u, v, w as functions of x and y, evaluated on a planar grid; ranges and samples of x and y	Vortex, Radial, Saddle, Helical
Formula streamlines	a *steady* field u, v, w; ranges of x and y and a number of seeds	Presets of the vector fields
Formula pathlines	a time-dependent field u, v, w(x, y, z, t)	Pulsating Vortex, Double Gyre, Travelling Wave, Oscillating Updraft (chapter 14)

13.2 Writing expressions

Element	Available
Operators	`+ - * / ^` (power; `2^3^2` is `2^9`), unary minus, parentheses
Functions	`sin cos tan asin acos atan sinh cosh tanh exp log sqrt abs sign` (one argument; `log` is natural and needs a positive value, `sqrt` a non-negative one), and the two-argument `pow(a, b) min(a, b) max(a, b)`
Constants	`pi`, `e`
Variables	`x`, `y` (and `z` where a third coordinate exists); `u`, `v` are the surface parameters of a parametric surface, `t` the parameter of a curve or, for pathlines, the *time*

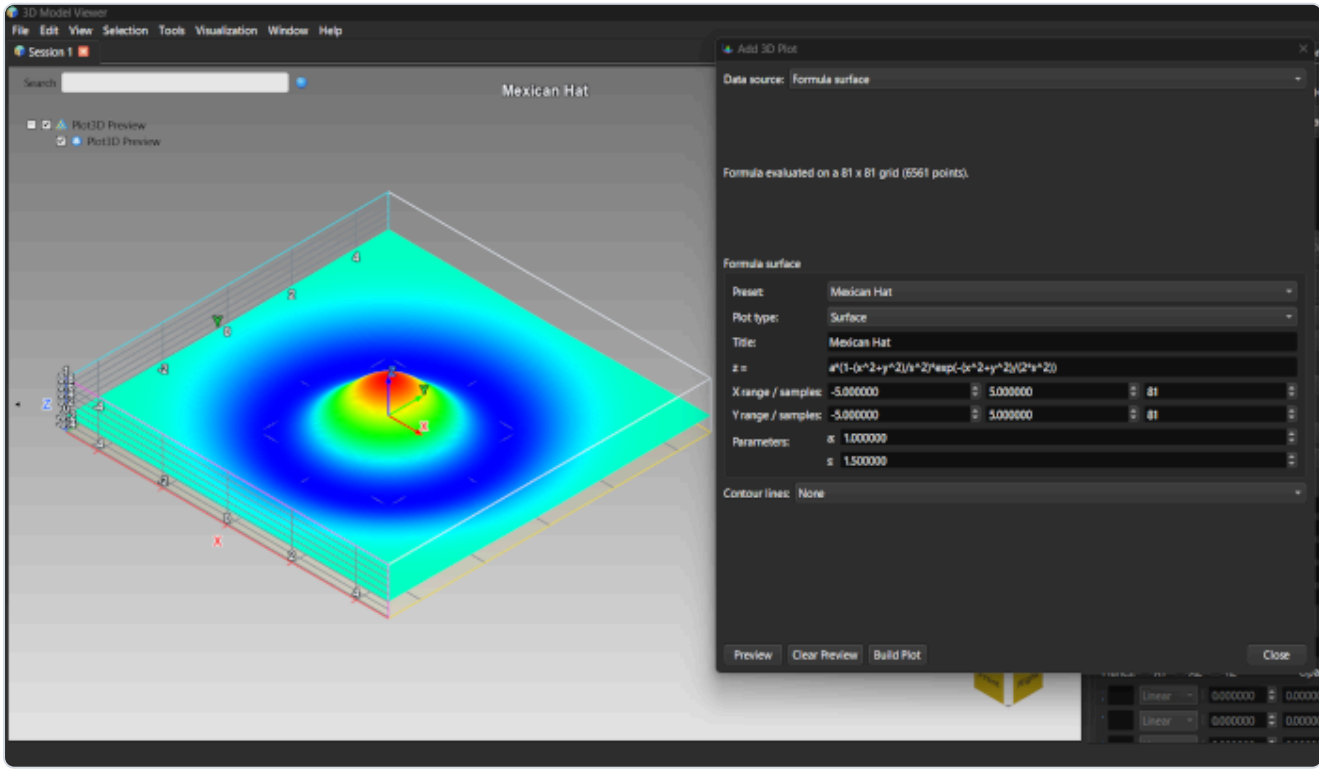
Element	Available
Parameters	Named constants such as `a`, `s`, `kx` that a preset provides and that you can edit - change one and press **Refresh Preview**

A mistake is reported in the status line under the preview, by name: *Unknown variable: q*, *Missing closing parenthesis*, *Division by zero*, *log requires a positive value*. **Build Plot** stays disabled until the expression evaluates. Examples:

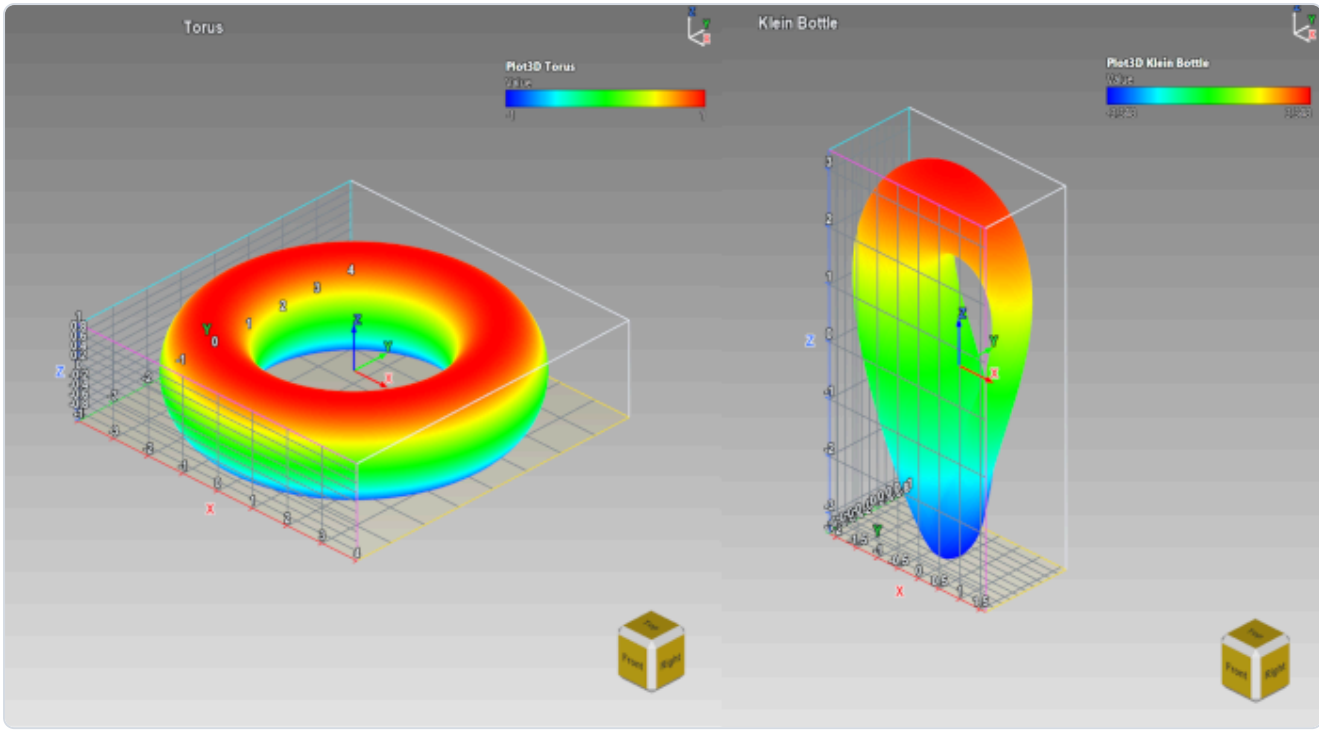
```
Formula surface   z =  a*exp(-((x-x0)^2+(y-y0)^2)/(2*s^2))       (Gaussian preset)
Parametric curve  x = cos(t)   y = sin(t)   z = t/(2*pi)          (a helix)
Implicit surface  f = x^2 + y^2 + z^2 - r^2                       (a sphere of radius r)
```

13.3 Preview, build and edit

Refresh Preview evaluates the formula and shows the plot in the scene, with a status line such as *Pathlines from 12 seeds over t = 0 … 10 (n segments), coloured by time*. **Build Plot** adds it. Later, select the plot in **Active plot**, click **Edit Plot…**: the dialog reopens on the same source with the expressions, ranges, sample counts and parameters restored; change one and click **Rebuild Plot**. The plot updates **in place**, keeping its title, visibility, colours and axis settings. The source type cannot be changed in an edit; build a new plot for that.



The Mexican Hat formula surface with the dialog and its status line



A torus and a Klein bottle from the parametric presets

TIP

Generated plots are a fast way to check a data pipeline or to build a picture with no data at hand: a Gaussian bump as a target, a saddle to show a minimum-maximum, a torus as a test object for the surface tools.

14 Flow in time: pathlines

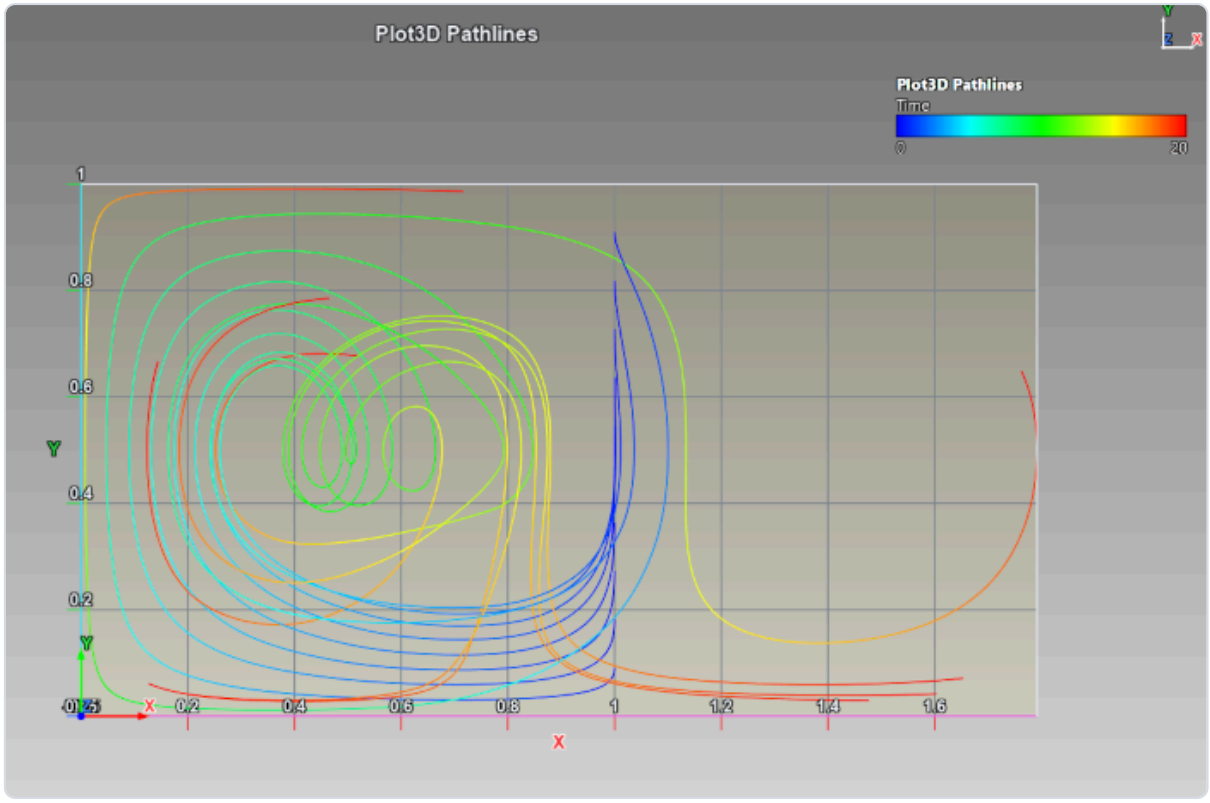
A **streamline** is the path along a field frozen at one moment. A **pathline** is the track of a particle carried by a field that *changes with time* - it is what a drop of dye would show. When the flow is unsteady the two differ, and pathlines are the ones that match reality. ModelViewer draws pathlines from a time-dependent formula or from a CSV time series, colours them by time, and can animate them.

14.1 Pathlines from a CSV time series

The table is a velocity field sampled on a **complete regular grid** for each of a series of times. Each row holds a time, a position and a velocity:

```
t,x,y,u,v,w
0,0,0,-0.000000,0.000000,0
0,0,0.25,-0.000000,0.222144,0
...
```

- Choose **Data source** *CSV time series (pathlines)* and open or paste the table. **Time column**, **X**, **Y**, an optional **Z** and **U**, **V**, **W** are guessed from the header names (`t` or `time` , `x` , `u` ...) and can be changed. Without a Z column the field is planar.
- The status line reports either *N rows form a complete grid* and how many seeds and segments result, or why the table cannot be used: a missing row (it says how many rows are needed), a non-numeric cell (it names the row), a column left out.
- **How it is traced.** Seeds are released along Y at the middle of the X range (in the middle z plane for a 3-D table) and carried through the field with a fourth-order Runge-Kutta integration, over **time steps** equal steps from the first to the last time in the table. The field is interpolated linearly in space and in time. A trail ends where its particle leaves the grid. The row order of the file does not matter.
- The trails are lines coloured by **time**, so the legend runs from the start to the end time, and they take the usual colour map, bands, range and line-width controls.



Pathlines of the double gyre, coloured by time

Sample	What it is
`pathlines_double_gyre.csv`	The classic unsteady double gyre on x 0..2, y 0..1: two counter-rotating cells whose dividing line oscillates with a period of 10 s, sampled at 21 times (945 rows). A planar field.
`pathlines_rising_vortex_3d.csv`	A 3-D field (4,212 rows with a Z column): a rotation about the Z axis whose rate pulses in time plus an oscillating vertical velocity, so the trails climb as they circle.
`pathlines_openfoam_cavity.csv`	The cell-centre velocity of the OpenFOAM cavity sample (times 0 to 2 s) on a plane just above its top face, made to be played together with that result (chapter 18).

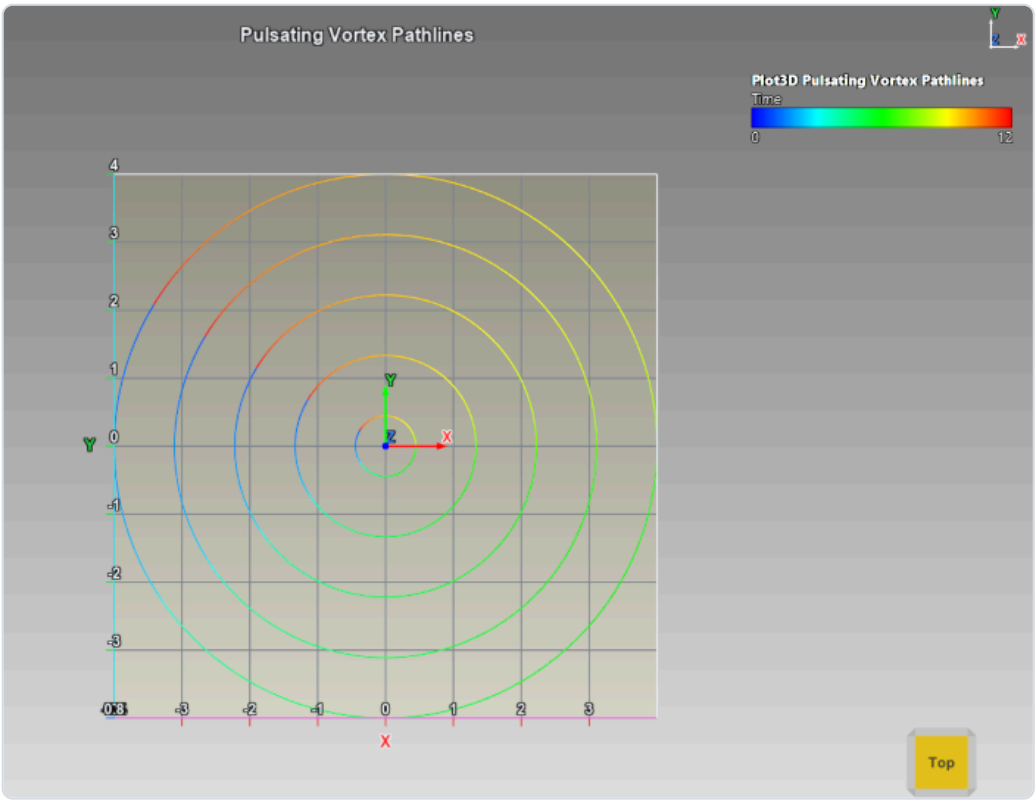
14.2 Pathlines from a formula

With **Formula pathlines (time-dependent)** the field is three expressions **u(x,y,z,t)**, **v(x,y,z,t)**, **w(x,y,z,t)** in position **and time**; **Y range / samples** sets the seeds along Y at the middle of the X range (at z = 0), and **T range / steps** the time interval and the number of integration steps. A trail ends where its particle leaves the x / y range.

Preset	Behaviour
Pulsating Vortex	A rigid rotation whose rate pulses in time: concentric rings; the colour (time) bands bunch up where the particle slows and stretch where it speeds up.
Double Gyre	Two counter-rotating cells whose dividing line oscillates; neighbouring trails fold and stretch.

Preset	Behaviour
Travelling Wave	A steady stream deflected sideways by a wave travelling in x: the trails drift forward and wiggle.
Oscillating Updraft	A vortex with a vertical velocity that oscillates: the rings climb and sink, so the trails draw helices in Z.

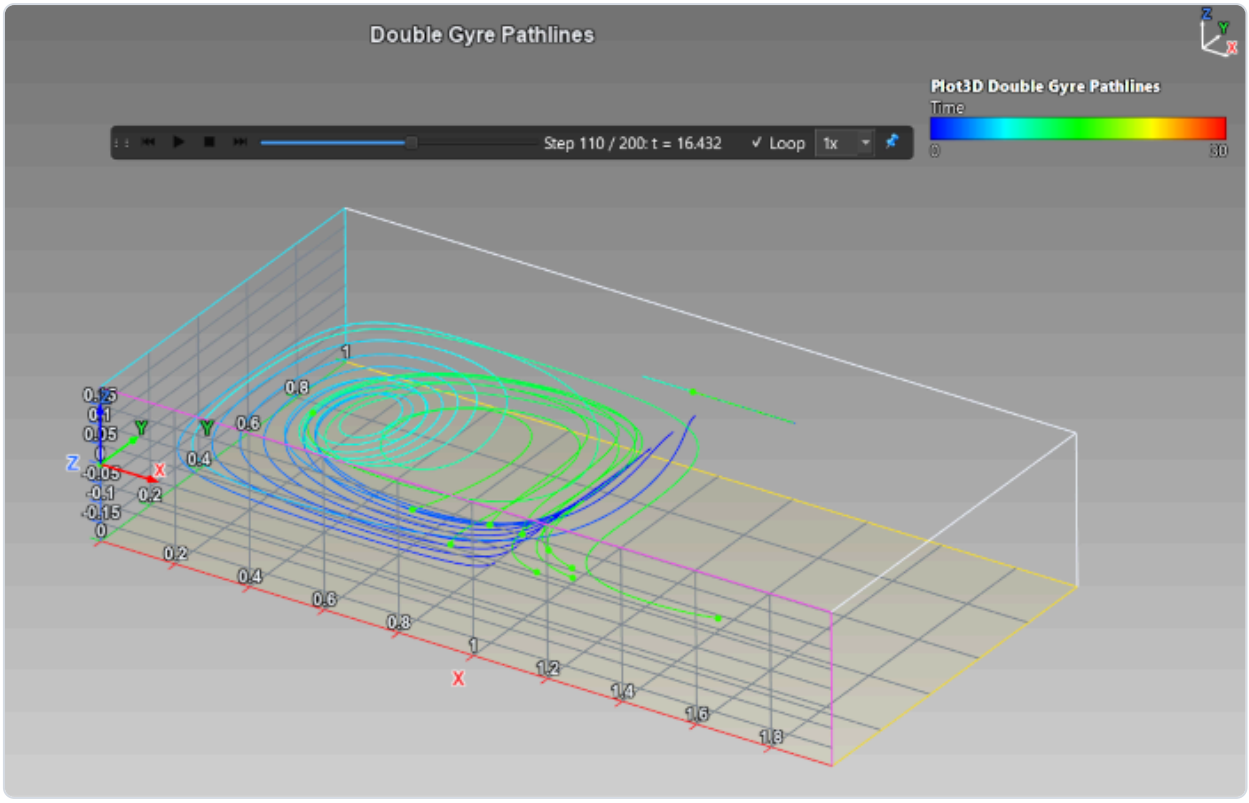
> **CAUTION**
> In this source `t` is **time**. In *Formula streamlines* the field is steady (no time), and in *Parametric curve* `t` is merely the curve parameter. If the end of the time range is before its start, **Build Plot** is disabled with a message.



The Pulsating Vortex preset: rings whose colour bands bunch up where the particle slows

14.3 Animating pathlines

Select the plot in **Active plot** and tick **Animate pathlines (timeline)** in the 3D Plot tab. A **timeline** appears at the top of the viewport - the same bar as for simulation results, with play/pause, step, stop, a slider, the current time, **Loop** and a speed choice. The trails are drawn only up to the current time, with a dot at the head of each, coloured by the current time. Switching the option on shows the whole plot until you press play; from the end, play starts over; unticking the option brings back the full trails and removes the bar. Hiding the plot in the scene tree removes the bar and pauses playback.



Pathlines part-way through the animation, with their head dots and the timeline

The animation is a viewing aid: it is not saved with the file and the head dots are not exported.

15 Styling, axes and scales

Everything about a built plot - colours, sizes, the axes - is in the **3D Plot** tab of the dock (next to the Simulation Results tab). Choose the plot in **Active plot**; the controls act on it. The plot's entry in the scene tree has the visibility checkbox.

15.1 Colour

Control	What it does
Title	The name shown in the scene tree and the legend.
Colour map	The colour ramp: sequential, diverging, and others.
Colour bands	Smooth colours, or a number of discrete bands.
Automatic colour range	On: the range follows the data (also after an edit). Off: **Minimum** and **Maximum** are yours - use it to compare plots.

15.2 Size and shape

Control	For
Line width	lines and pathlines (in screen pixels)
Marker size	scatter markers
Arrow size	vector fields (length scale; does not grow with the camera)
Bar width, **Bar depth**	bars
Contour levels, **Contour lines**	surfaces: how many iso-lines, and whether on the surface or on the base plane
Opacity	image planes and the reference planes

15.3 The axes box and the reference planes

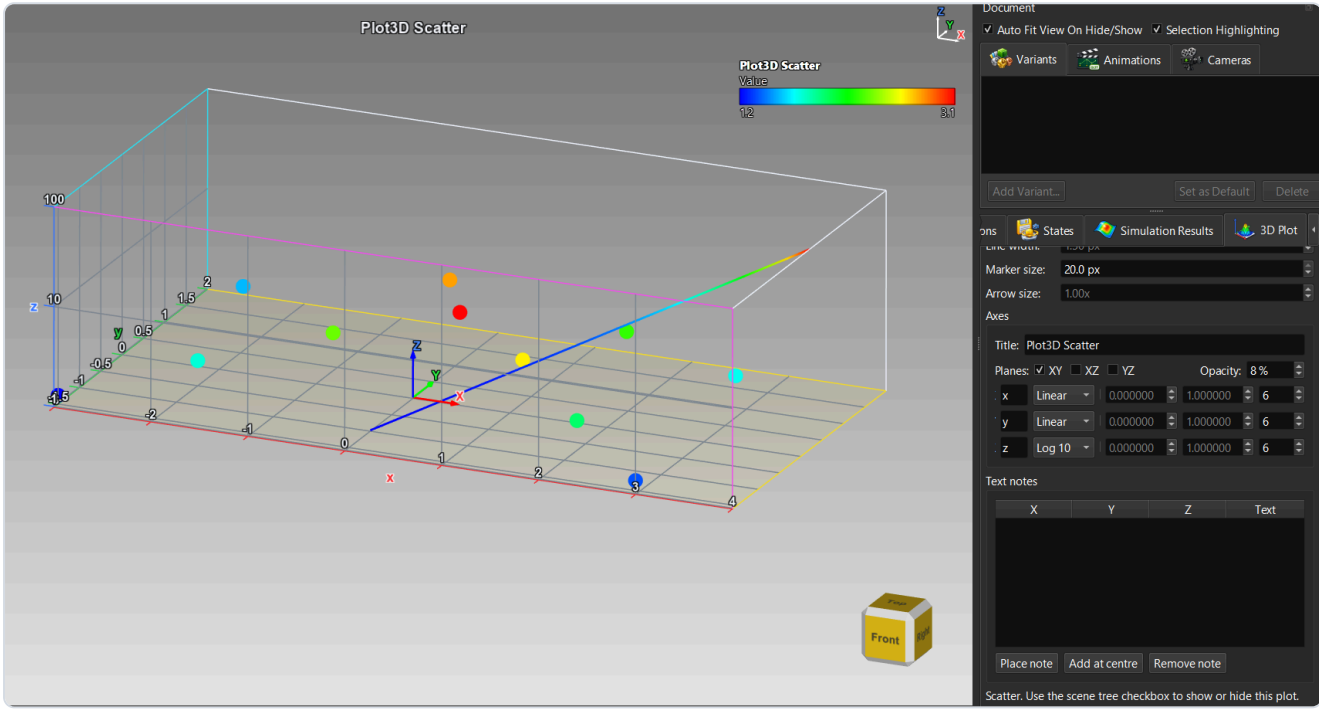
Show axes box frames the plot with labelled axes, ticks and numbers; hiding it does not discard the layout. **Planes** adds the reference planes XY, XZ and YZ at the minimum of the other axes, with an **Opacity**. A flat axis (all values equal) is padded by a tenth of the largest other extent, so a plane does not turn into a needle.

15.4 Shared axes

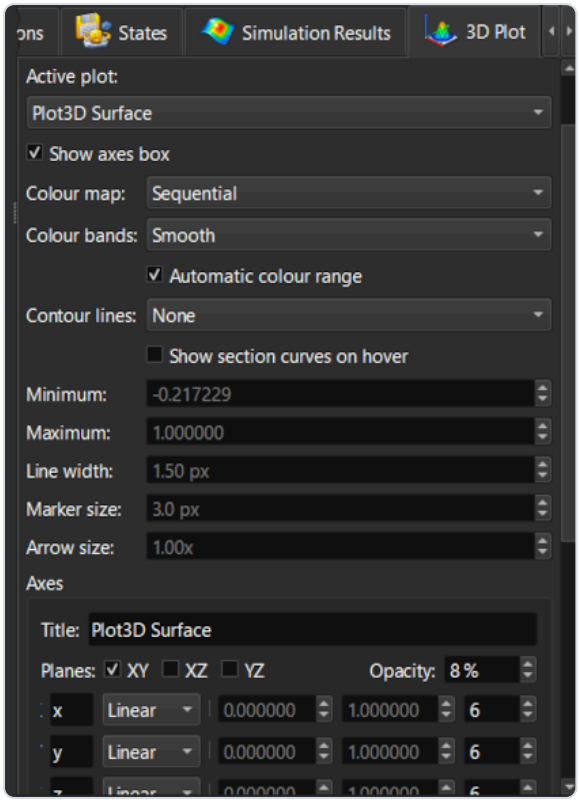
All plots in a document are drawn on **one set of axes** - the same scales, ranges and labels - because that is what lets you compare them. Under **Axes** every axis has a label, an **Axis scale**, a range and a tick count. Changing one re-lays out **all the plots together**, whichever plot is selected; a plot built later joins the axes that are already there.

Scale	Meaning and limit
Linear	The default.
Log 10	Powers of ten. Needs the data of *every* plot to be above zero on that axis. Setting it while a plot reaches zero is refused with a message naming the reason; building a plot that cannot be shown on an existing Log 10 axis returns that axis to Linear for all plots, with a status message.
SymLog	Logarithmic for large values, linear around zero; works for any data including negative values. Voxel indices that start at 0 and vector fields with both signs are examples.

- Lines, scatter, surfaces, bars, vector arrows and voxel volumes all follow the scale: the geometry is transformed, not just relabelled.
- Vector arrows are re-aimed by the local stretch of the scale at their site and keep their length (it carries the magnitude) and colour; voxel grids are resampled onto a grid that is regular in the scaled space. Returning to Linear restores everything.
- The hover readout of a surface still shows the plot's **data values**, not positions on the scaled axes.



A scatter plot and an exponential line sharing a logarithmic Z axis



The 3D Plot tab with the controls of the active plot

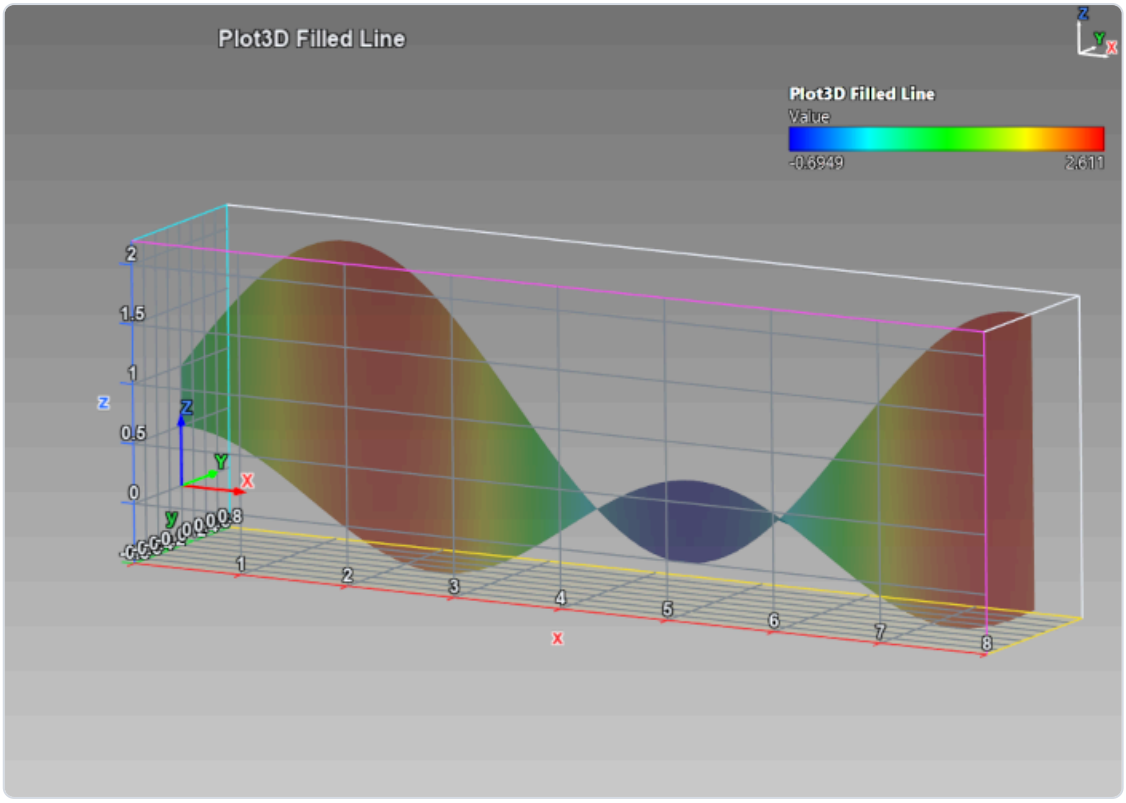
16 Fills, notes, images and probing

16.1 Fills: ribbons and curtains

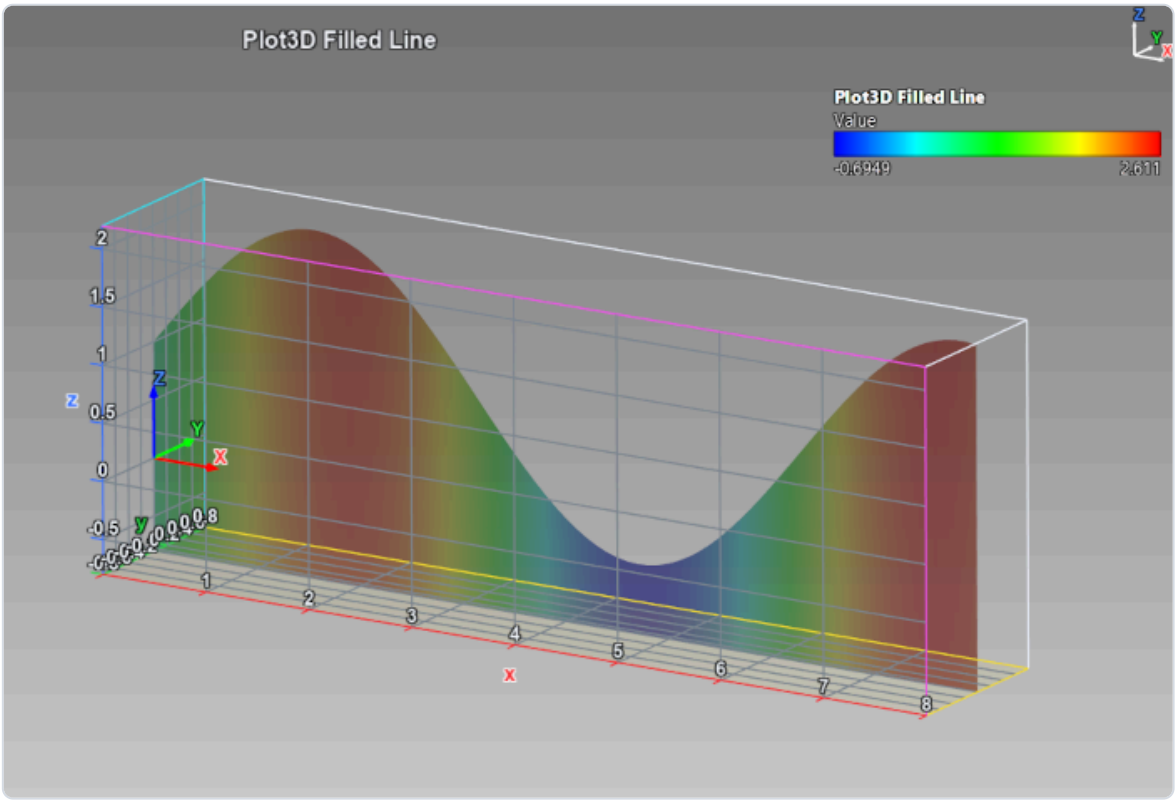
Lines and scatter plots can be filled with a translucent ribbon, coloured like the plot, to a base plane or between two curves.

Goal	How
A curtain under a line	Line / Curve (`line_helix.csv`): tick **Fill to Base Z** and set the base (0). Edit Plot to change the base; the curtain hangs from the line where it lies below the base.
A band between two curves	Line / Curve with a second Z column (`line_band.csv` : columns x, y, z, z2, value): tick the fill and choose the column in **or between the line and column**. The band pinches to a line where the curves cross. The base box is unused while a column is chosen; with the column set to *(none)* the fill reaches the base plane again.
Ribbons from scatter points	Scatter (`scatter_clusters.csv`): **Fill to Base Z** with a base such as -3; or a column in **or between the plot and column** (`scatter_band.csv`) for ribbons between two sets of points.
Fill a generated curve	Parametric curve, Helix preset: **Fill the curve down to Base Z**, or type a second z2(t) (for example `2 + sin(t)`) in **or up to z2(t) =** to fill between two curves with the same x(t), y(t). Whether a generated plot is filled is decided when it is built.

The axes box always grows to include what the fill reaches.



A translucent band between two crossing curves, pinching to a line where they cross

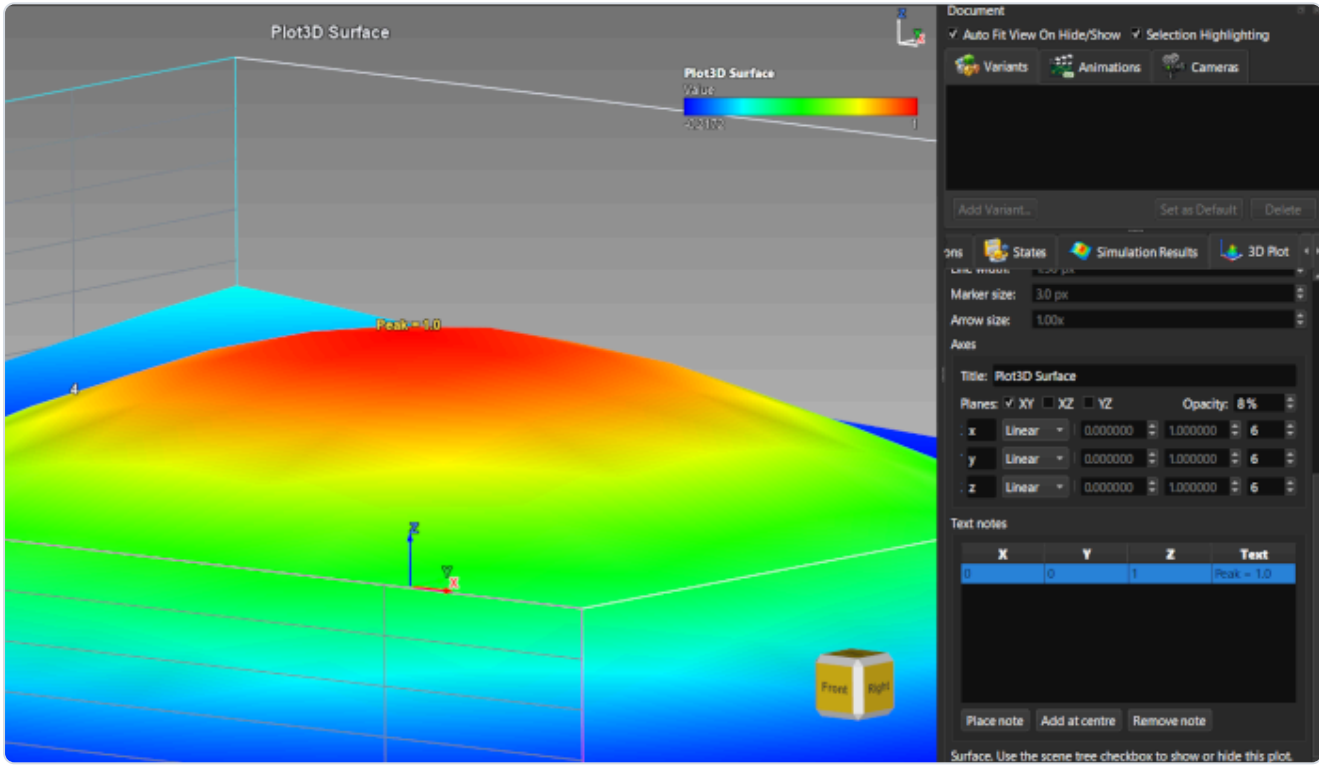


The helix with a curtain down to the base plane

16.2 Text notes

A **note** is a short text at a point of your data - a peak, an outlier, a threshold. Select a plot and open the **Text notes** group in the 3D Plot tab.

- **Place note** arms a click tool: click a point on the plot. The click snaps to the nearest data point of a visible plot (within 14 pixels; for lines, also the closest point on a segment), otherwise it uses the surface point under the cursor. A box asks for the text, which may have several lines. **Add at centre** adds a note at the centre of the plot's data.
- In the view, **drag** a note to move it, **double-click** to edit it, **right-click** to delete it. The table in the tab edits the note's X, Y, Z and text directly.
- Notes are stored in the plot's *data* coordinates, so they stay on their point when the axis scale changes. They follow the plot's visibility, are drawn even when the axes box is hidden, and are saved with the session.
- Every placement, move, edit and deletion can be **undone**.



A note on the peak of a surface

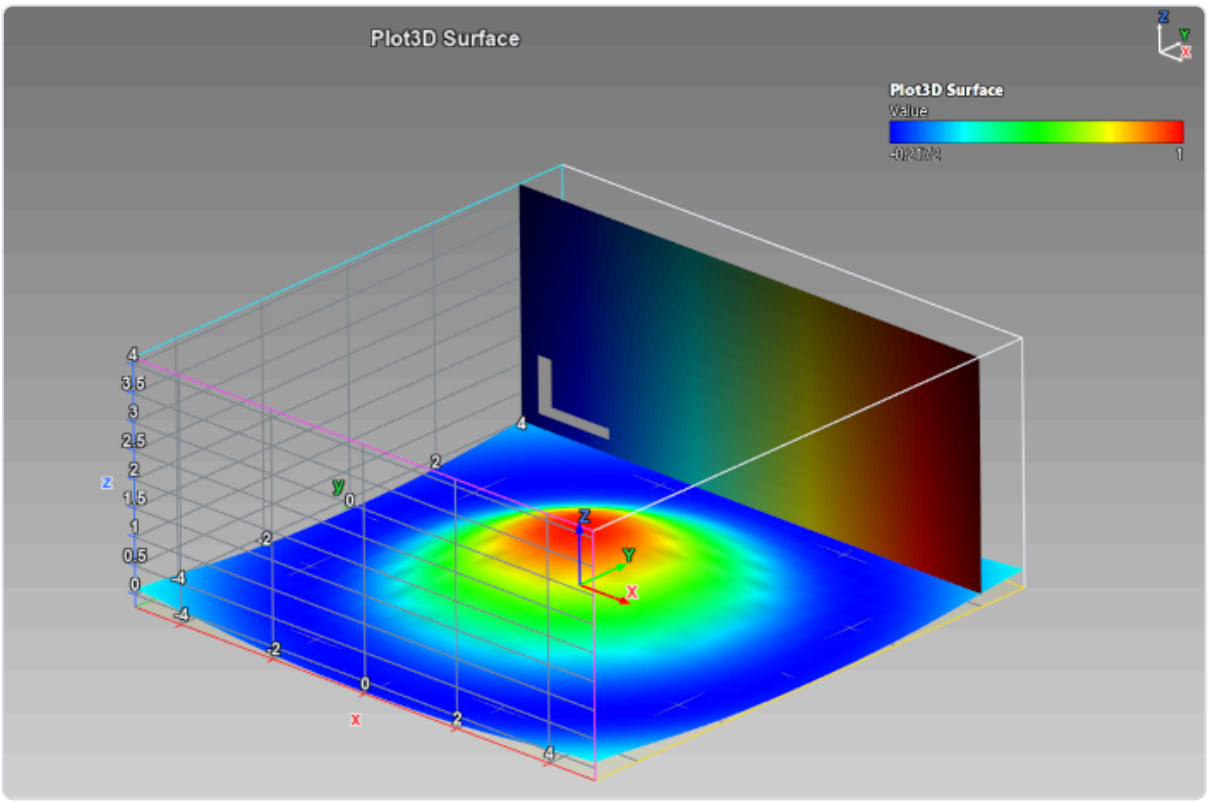
16.3 Section curves on hover

For a **surface**, tick **Show section curves on hover**. Moving the cursor over the surface draws, through the hovered point, the curves where the X, Y and Z planes cut it - coloured red, green and blue like the axes (the blue one is the contour line through the point) - together with the point's data values. Only the connected curve through the point is drawn, so a saddle shows one branch. It is a quick way to read a profile of a surface without building a chart, and it works in the single view only.

16.4 Images on a plane

An **image** can stand on one of the plot planes: a site map under a surface, a photograph of a specimen beside its measured data. Choose **Data source** *Image on a plane*, pick the file, and choose the plane: **XY** (flat, at the Z minimum), **XZ** or **YZ** (upright, at the minimum of the remaining axis); the ranges adapt to the picture's shape when you pick it or change the plane, and you can adjust them.

- **Opacity** (100 % by default) lets what is behind the picture show through; a picture with its own transparency combines with it (`image_gradient_alpha.png` fades towards its corners).
- Seen from behind, a picture reads **mirrored**. For an opaque picture, tick **Readable from behind** and the back shows it the right way round (`image_gradient.png` has an "L" mark to show which way round it is). The option is available only for an opaque picture at 100 % opacity - with a transparent area or a lower opacity both sides would show through each other, so the back stays mirrored.
- An image plot takes no colour map or legend: its pixels are its colour.



An image standing on a plane next to a surface plot

17 Editing, saving and exporting

17.1 Editing a plot

Every plot remembers how it was made. Select it in **Active plot** and click **Edit Plot...**: the dialog reopens on the same source with the table, the column choices, the expressions, the ranges, the sample counts and the parameters restored. Change what you need and click **Rebuild Plot**: **the plot is rebuilt in place**, with no new tree entry, and keeps its title, visibility, colour map, range and bands, reference planes, contour lines, line width and axes settings (with **Automatic colour range** on, the range follows the new data). What cannot change in an edit is the data source (and, for a formula surface, the surface/contour choice); build a new plot for that.

17.2 Saving

Plots are saved in the `.mvf` session with everything needed to edit them later: the data, the definition (columns or expressions), the axes, the colour settings and the notes. After reopening, **Edit Plot...** still works. A session saved by an older version that had different axes for each plot opens on the selected plot's axes, now shared.

17.3 Exporting

- **Point and line plots** (scatter markers, lines, pathline trails) export to **glTF**, **GLB** and **OBJ** with the rest of the scene; their colours are baked into the file so other programs show them coloured.
- Surfaces, bars and other mesh plots export like any mesh.
- **Images** of the view - for reports, slides, papers - come from the program's image export and the PDF report, like any other view.
- Plots are **not drawn by the ray tracer**, and path-traced renders do not contain them. The animation of pathlines and the notes' on-screen drawing are viewing aids and are not part of an exported model.

> **TIP**
> To get your numbers back out, keep the CSV: a plot's definition is saved, but the original table remains the master copy. If you generate data from a script, rerun it, reopen the plot with **Edit Plot...** and load the new table.

PART III

Together

18 Results and plots on one clock

A simulation result and a plot live in the same scene, so they can be shown together. When both change over time, one **playback bar** plays them in step. Which combination makes sense depends on whether the data shares coordinates and time.

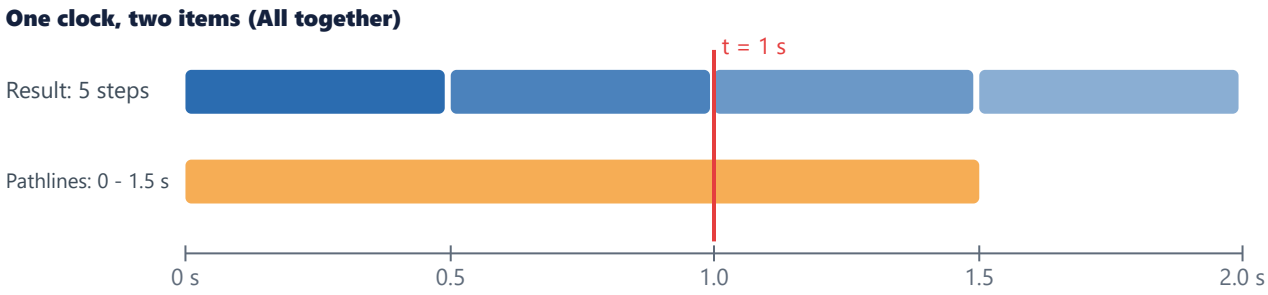
18.1 Four situations

You have	Same coordinates?	What is synchronised, and how it is shown
A transient flow or thermal result and pathlines of the same flow (for instance a CSV time series exported from the solver)	Yes	**Physical time.** One view, overlaid: the result's time equals the pathlines' time. Use *All together* on the playback bar (below).
A result and a chart of a point's history, maximum stress against time, a reaction force, or measured test data	No (a curve is not in model space)	**Physical time**, with a moving cursor on the chart and click-to-seek (chapter 7). The model and the chart are shown beside each other.
A modal analysis and a frequency-response curve	No	**Frequency.** The chart's cursor sits at the current mode's frequency; clicking a peak selects the mode.
A load-step or parametric sweep and a response curve	No	**The step's value** against the chart's axis.

> **THE RULE BEHIND IT**
> Data is matched by **physical value, not by progress**. A result from 0 to 2 s and pathlines from 0 to 1.5 s line up at 1 s. Matching by progress (0 to 100 %) is only the fallback when the units are incompatible - a modal result is in hertz and pathlines are in seconds - and the playback bar then says *by progress*.

18.2 The playback bar for both

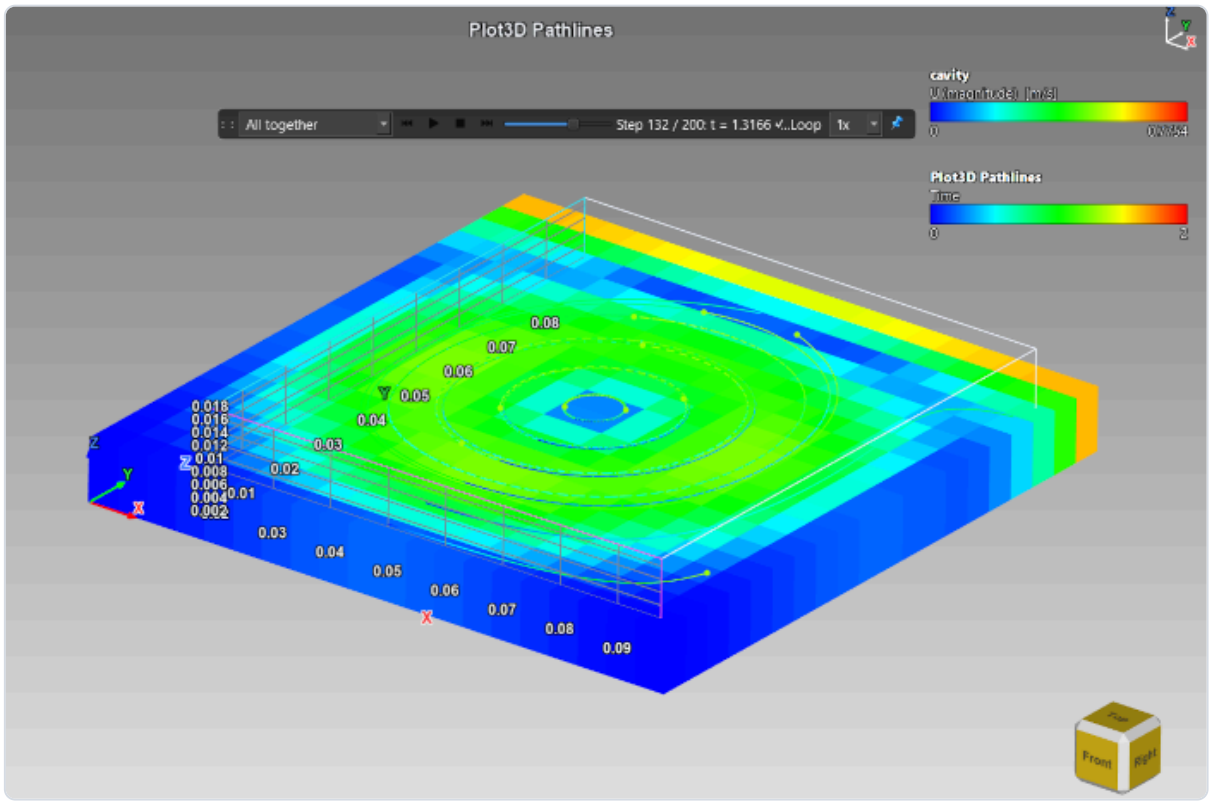
The timeline plays whichever item is chosen in the box at its left. The box lists every multi-step simulation result (*Simulation: name*), the animated pathline plot (*Pathlines: name*) and, when both can play, **All together**. With only one playable item the box is hidden. The bar follows what you work on: choosing a result in the Simulation Results tab selects it, and ticking **Animate pathlines (timeline)** selects that plot. Changing the selected item pauses the one that was playing. The legends of a result and a plot stack in one column, each headed with its owner's name.



Both items follow real time. The result holds each step until the time of the next; the pathlines hold their last frame after 1.5 s

18.3 Worked example: the cavity flow

1. Open `sample-models/Simulation/openfoam_cavity/cavity.foam` and choose **U** with the *Magnitude* component.
2. In the 3D Plot tab choose **Add 3D Plot…**, source *CSV time series (pathlines)*, open `sample-models/Plot3D/pathlines_openfoam_cavity.csv` and map Time = t, X = x, Y = y, Z = z, U = u, V = v, W = w. This file holds the cell-centre velocity of the same case on a plane just above its top face, so the two share coordinates and real time. Build the plot; hide its axes box if it clutters the view.
3. Tick **Animate pathlines (timeline)**, choose **All together** in the playback box and press play. The trails grow while the field steps from 0 to 2 s; the bar reads the time and the step, for example *t = 1 – step 3 of 5*.



The cavity result and its pathlines playing together

> **CAUTION**
> A modal result such as `FEM_box_modes.frd` has steps in hertz and cannot share a time axis with pathlines in seconds; **All together** then plays them by progress, and says so.

18.4 Charts as the second half

For the other three situations the chart does the work: plot a point's history (right-click → **Plot Over Time Here**), add your measured curve from a CSV, and play. The cursor travels along both curves; click in the chart to jump in time. For a modal result add a frequency-response CSV and click a peak to jump to that mode. Chapter 7 has the details and the sample files.

19 Worked workflows

Recipes that combine the chapters. Each lists the sample files to try it with.

19.1 Review a structural analysis

1. Open the result (`plate_with_hole.frd`), look at the info line, and set the units under **Values are in**: this file is in mm and MPa.
2. Colour by **von Mises**, keep *Automatic (all steps)*, tick **Mark minimum and maximum**.
3. Switch on **Show deformed shape** with Auto scale, and play the four load steps.
4. Turn on **Show stress ellipsoids** (count 1500+) to see the stress concentration at the hole.
5. Use **Plot Over Line** from the edge of the hole to the plate edge to read how the stress falls off, and save the chart image.
6. Place two annotations (**Tools** → **Annotate...**) at the maximum, capture views in the Cameras panel and export the report (**Tools** → **Export Report...**).

19.2 A thermal transient against measurements

1. Open `FEM_box_thermal_transient.frd` ; the file states no temperature unit, so choose one under **Values are in**.
2. Right-click the middle of the far face → **Plot Over Time Here**.
3. In the chart, right-click → **Add curve from CSV...** → `thermal_probe_test.csv` . Play the timeline and compare the curves.
4. Add a second point (**Add a point from the model**) near the heated face to see the heat arrive earlier. Save the chart as an image.

19.3 A flow with its particle tracks

Follow the cavity example of chapter 18. For a steady 3-D flow without a time series, use streamlines of the result instead: `duct.cgns` with **Show streamlines** (Velocity), **Seed on the Clipping Plane** and a clipping plane across the duct.

19.4 Modes and a frequency response

1. Open `FEM_box_modes.frd` , switch on the deformed shape, and step through the six modes.
2. Right-click a point → **Plot Over Time Here**: the x axis is the mode frequency.
3. Add `FEM_box_modes_frf.csv` and click a peak: the result jumps to the corresponding mode.

19.5 Two design variants

1. Open the first result, add the second with **Add Result to This Document...**.

2. Compare side by side with **Same colour range for both** and **Link the cameras** (`ibeam_cantilever.frd` against `ibeam_torsion.frd`).
3. Exit compare and chart the same point of each result; export both charts as CSV and load one into the other's chart to see the two curves together.

19.6 Measured data next to a CAD model

1. Open the CAD part, then add a scatter plot of your measurements (columns x, y, z and a deviation as the colour value); both are in the same coordinates if the table uses the model's units.
2. Set the axes to match the model, add notes at points of interest, and hide the axes box if you only want the points.
3. For a picture of the specimen, add an *Image on a plane* at the right place.

> **TIP**
> Save the document as `.mvf` at each stage worth keeping (chapter 9 explains what a saved result contains): a session with results, plots, annotations and captured views is the best single record of an analysis review.

A Sample file catalogue

The samples ship in `sample-models/Simulation` and `sample-models/Plot3D`; each folder has a `README.txt` that describes every file and how it was made. Values in the generated files are synthetic: they test the program, not physics.

A.1 Simulation results

File	What it is	Good for
`FEM_box_static_stress.vtu`	Static analysis of a 10 mm box (CalculiX via FreeCAD): 280 nodes, 129 quadratic tetrahedra; displacement, von Mises, Tresca, principal stresses	First look, sections
`FEM_box_frequency_mode1.vtu`	The first vibration mode of the same box	Modal display
`FEM_box_static.frd`, `FEM_box_frequency.frd`	The same analyses as the solver wrote them (mm, MPa); von Mises etc. are derived on load	Units, derived fields
`FEM_box_load_steps.frd`	A load ramped in four increments	Timeline, fixed colour scale
`FEM_box_modes.frd`, `FEM_box_modes_frf.csv`	Six vibration modes; a synthetic frequency-response curve with six peaks	Modes, chart seek
`FEM_box_thermal_transient.frd`, `thermal_probe_test.csv`	Transient heat transfer, 20 steps over 10 s; synthetic thermocouple readings	Time chart, CSV curve, units
`plate_with_hole.frd`	Plate with a hole, four load steps (CalculiX, gmsh mesh)	Stress concentration, ellipsoids, chart over line
`ibeam_cantilever.frd`, `ibeam_torsion.frd`	An I-beam in bending and in torsion, four load steps each	Compare, ellipsoids
`stress_states.vtu`	Block with three stress tensors (bending, torsion, element-wise)	The Tensor field choice
`cell_data_cube.vtk`	8 x 8 x 8 hexahedra with cell data only (*Element_Stress*, *Element_Group*)	Cell data, averaging
`shell_plate_transient.frd`, `plate.vtk`	A shell plate with 10 temperature steps; a vibrating plate with several vector fields (surface only)	Shell charts
`openfoam_cavity/cavity.foam`	A 20 x 20 x 1 cell OpenFOAM case, five time directories (synthetic fields U, p, T, sigma)	Cell data, units from dimensions, pathlines
`block.cgns`, `block.exo`, `block.vtkhdf`	The same 8 x 8 x 8 block in three formats, five steps: Temperature, Velocity (vector), a per-cell quality	Iso-surfaces, volume, format readers
`duct.cgns`	A half-ring duct of two structured blocks, four steps of a swirling flow	Streamlines

File	What it is	Good for
`pointe_4fields.med`, `fra.med`	MED files from the Salome sample set (mixed cell types; a nodal velocity and void fraction)	MED reader
`polyhedra.*`, `hexa.vtk`, `post.vtk`, `tetraMesh.vtk`, `uGridEx.vtk`, `beampl.frd`	Reader tests: polyhedra, a smooth scalar, a pressure field, a tiny tetrahedral mesh, mixed cell types, a CalculiX example	Format tests
`real/` and `large/`	Files written by other tools (VTK test data) to test the readers; a generator script for very large results (1 million cells; not in git)	Reader and lazy-loading tests

A.2 3D plots

File	What it is	Plot type
`surface_ripple.csv`	sin(r)/r on a 21 x 21 grid	Surface, Contour
`surface_scattered.csv`	Fourteen irregular points with a sloped height	Surface (triangulated)
`line_helix.csv`	200 points of a two-turn helix	Line / Curve, fill
`line_band.csv`	Two crossing curves (columns z and z2) coloured by the band width	Line, fill between curves
`line_growth.csv`	z = 10^(x/2), from 1 to 100	Line, log axis
`scatter_clusters.csv`	Three clusters of 60 points, coloured by id	Scatter, fill
`scatter_error_bars.csv`	Points with an error column	Scatter, error bars
`scatter_band.csv`	Two sets of Z values (z and z2)	Scatter, fill between
`bar_histogram.csv`, `bar_histogram_1d.csv`, `bar_signed.csv`	An 8 x 8 bump of bars; eight bin counts; bars above and below their base	Bar / Histogram
`voxel_sphere.csv`	A sphere in a 9 x 9 x 9 occupancy grid	Voxel / Volumetric
`quiver_vortex.csv`	A rotation about Z with an upward drift (7 x 7 x 3 grid)	Vector (Quiver)
`pathlines_double_gyre.csv`, `pathlines_rising_vortex_3d.csv`, `pathlines_openfoam_cavity.csv`	Time-dependent velocity fields on regular grids	CSV time series (pathlines)
`image_gradient.png`, `image_gradient_alpha.png`	A 320 x 200 picture with an "L" mark; the same with fading transparency	Image on a plane

B Quick reference

B.1 Where things are

To do this	Go to
Open a result in its own document	**File** → **Open...**
Add a result to the current document	**Visualization** → **Simulation Results** → **Add Result to This Document...**
Compare two results	**Visualization** → **Simulation Results** → **Compare Results...** or the tab's **Compare**
Change field, range, colormap, deformation, sections, volume, arrows, ellipsoids	The **Simulation Results** tab (bottom-left dock)
Quick history of a point	Right-click the model → **Plot Over Time Here**
Add a 3D plot	**Visualization** → **Plot 3D...** or the **3D Plot** tab → **Add 3D Plot...**
Style a plot, set the axes, notes, edit, animate	The **3D Plot** tab
Sections through a result	The Section View button of the View toolbar
Annotate, measure, report	**Tools** → **Annotate...**, **Tools** → **Measure...**, **Tools** → **Export Report...**
What is new, the tutorial, quick help	**Help** → **What's New...**, **Help** → **Tutorial**, **Help** → **Quick Help**

B.2 Mouse and keyboard

Where and what	Action
Viewport: rotate, pan, zoom	Middle button drag, right button drag, wheel (see the tutorial, lesson 3)
Hover over a result	Shows the value under the cursor
Chart: zoom	Wheel; Shift +wheel x only; Ctrl +wheel y only
Chart: pan, reset	Middle button drag; double-click
Time chart: move in time	Click or drag in the chart
Opacity curve (volume)	Drag a point; double-click adds; right-click removes an inner point
Timeline	Play/pause, previous/next, stop, slider, Loop, speed; drag the bar to move it, double-click to reset it
Plot notes	Place note, then click; drag to move, double-click to edit, right-click to delete

B.3 Limits at a glance

Quantity	Value
Iso-surface levels	1 to 20
Streamline seeds	1 to 500 (2000 steps or four model diagonals per direction)
Volume quality	Low 32, Medium 64, High 96, Very high 128
Ellipsoid count	20 to 5000
Time steps stored in a session	100 per result (evenly spaced)
Results read lazily above	256 MB of step data, with more than one step

C Messages and troubleshooting

C.1 What you may see, and what to do

Situation	Cause and remedy
A format is missing from the Open dialog	Your build lacks the library for it (NetCDF for Exodus, CGNS, HDF5 for VTKHDF and MED). **Help** → **About** shows what is enabled.
"A simulation result is already being loaded"	Only one result loads at a time; wait for the progress bar or cancel it.
The result opens grey, with no colours	The file has no fields (only geometry). The panel says so.
Numbers have no unit in the legend	The file does not state one. Choose it under **Quantity** and **Values are in**.
Values look wrong by a factor of 1000 or 1,000,000	The unit of the file differs from what you assumed (Pa and MPa; mm and m). Check the units of the analysis and set them.
Colours differ between steps or between results	The range is per step or per result. Use *Automatic (all steps)*, a *Custom* range, or **Same colour range for both**.
"This result has no displacement field"	It cannot be shown deformed; the file has to contain displacements.
Sections, iso-surfaces, streamlines or volume do nothing	The result has no volume cells (a shell), or the selected field is not a node field (streamlines need a node vector field). The info line under the group says which.
Iso-surfaces are hidden	Switch on a clipping plane and switch off **Colour the Clipping Plane cut with the field**: an opaque cut hides what is behind it.
"Cut faces and iso-surfaces are shown as they were saved (frozen)"	The session was saved without the volume. Save with **Also store the volume** to keep them live.
No tensor to choose for the ellipsoids	The result has no six-component field with *stress* in its name.
A chart is empty or one dot	A one-step result shows its history as a point; a click that missed the result gives no data.
Compare button unavailable	The document needs two results; compare ends by itself if one is hidden or closed.
Plot: "Open or paste tabular data first"	The CSV has not been loaded.
Plot: a table that is not a complete grid	A surface or pathline table must have every grid combination once; the message says how many rows are needed.
Plot: "Unknown variable", "Division by zero", "log requires a positive value"	Fix the expression; **Build Plot** stays disabled until it evaluates everywhere in the range.
Plot: an axis went back to Linear	A Log 10 axis cannot show a plot whose values reach zero; every plot returned to Linear with it. Use SymLog, or remove that plot.

Situation	Cause and remedy
A picture appears mirrored	You see the back of an image plane. Tick **Readable from behind** (opaque pictures only) or look from the front.
Glass or other effects are missing in the plot	Plots and results are drawn flat in every mode and are not ray traced.

D Glossary

Boundary surface
The faces of a solid result that belong to exactly one cell: the skin that is drawn.

Cell data
A field with one value per element, drawn flat (also: element data).

Contour
A line (or band edge) of equal value.

Derived field
A value computed from another field, such as von Mises stress from a stress tensor.

Displacement
The vector by which a node moved under load; the basis of the deformed shape.

Field
A quantity defined over the model: scalar, vector or tensor.

Iso-surface
The surface in a volume where a field has one constant value.

Lazy loading
Reading a step from the file only when it is shown, so memory does not grow with the number of steps.

Mode shape
The pattern of movement of a structure vibrating at a natural frequency; its amplitude is arbitrary.

Node data
A field with one value per mesh point, interpolated smoothly.

Pathline
The track of a particle carried by a field that changes with time.

Principal stress
One of the three normal stresses in the orientation where the shear stress vanishes.

Quiver
A plot of arrows, one for each point of a vector field.

Snapshot
The data a saved session keeps of a result: the surface, the chosen fields and steps, units and view state.

Step
One state of a result: a load increment, a time step or a mode.

Streamline
The path along a vector field frozen at one moment.

SymLog
An axis scale that is logarithmic for large values and linear near zero, so it can show negative numbers.

Tensor glyph
: A symbol (here an ellipsoid) whose shape and size show a tensor.

Von Mises stress
: A single equivalent stress derived from the full stress tensor, used to judge yielding of ductile materials.